

CEOs' general managerial skills and corporate risk taking subject to the moderator of CEO tenure

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Abstract

In this study, we examine the relation between chief executive officers' (CEOs') general managerial skills and firms' risk-taking behaviour. We find that generalist CEOs are associated with significantly higher firm risk, with the association decreasing significantly with CEO tenure. We propose the following managerial skills transformation explanation: the longer a CEO stays with a firm, the less general and more firm specific the CEO's skills and knowledge become; therefore, any effect of the CEO's general managerial skills only appears in the early years of tenure.

KEYWORDS

CEO tenure, CEOs' general managerial skills, firm risk

JEL CLASSIFICATION

G32, J24

1 | INTRODUCTION

The general managerial skills of chief executive officers (CEOs) are presumably transferable across firms and industries and impact managerial behaviour and corporate decisions. Custódio et al. (2013) construct a general ability index (GAI) by considering the different aspects of a CEO's professional career, including the number of firms and industries in which the CEO has worked, prior CEO positions held at different companies, and experience with conglomerate

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firms. Using this index, many researchers have investigated the effects of CEOs' general managerial skills on CEO compensation (Brockman et al., 2016; Custódio et al., 2013; Liu et al., 2020), corporate strategic policies regarding innovation (Custódio et al., 2019) and mergers and acquisitions (M&As; see G. Chen et al., 2020), and firms' cost of capital (Mishra, 2014) and performance (Betzer et al., 2020; Gounopoulos & Pham, 2018).

A recent debate in the literature has focused on the role of CEOs' general managerial skills in firms' risk-taking behaviour. Custódio et al. (2013) contend that, since generalist CEOs have chosen a more diversified career, they are expected to be more risk averse than specialists. However, other scholars hold the opposite view, arguing that generalists are likely to take more risks than specialists. According to Custódio et al. (2019), Gounopoulos and Pham (2018) and Mishra (2014), general managerial abilities allow generalists to switch firms more easily; thus, their personal wealth is less contingent on their firms' performance and longevity than that of specialist CEOs. Therefore, generalists have stronger incentives to undertake risky projects. Brockman et al. (2016) further note that, since job switching causes CEOs to lose firm-specific skills, generalist CEOs who change jobs more frequently are likely to be less risk averse. Consistent with the notion of generalists being less risk averse, Mishra (2014) documents that firms run by generalist CEOs have a higher cost of equity capital, Gounopoulos and Pham (2018) report that generalist CEOs in newly listed firms are associated with a higher probability of failure and shorter survival time, and Custódio et al. (2019) find that generalist CEOs promote innovation.

Although previous studies have provided evidence showing the important role of CEOs' general managerial skills in firms' risk-taking behaviour, these studies have examined the overall effects on the risk-related corporate policies or firm-level outcomes of CEOs' general managerial skills without paying attention to the potential evolution of the effects with CEOs' tenure. Noting a potentially important feature of CEOs' general managerial skills, we expect CEOs' tenure to have a significant impact on the relation between CEOs' general managerial skills and firms' risk-taking behaviour. In particular, we emphasize the following "general to firm-specific" transformation process of general managerial skills that is intuitive but has not been noted in previous studies: as a CEO stays with the same firm, the CEO's skills gradually become less general and more firm specific. Intuitively, with the passage of time, a CEO's skills and knowledge of the firm's business and customer and supplier relationships pertaining to the industry become established and accumulate, whereas those unrelated to the firm and the industry gradually become obsolete. A strong empirical implication is that, if there is any effect of generalist CEOs on firms' risk-taking behaviour, this effect is only evident when CEOs are in their early tenure. In this study, we examine the relation between CEOs' general managerial skills and firm risk by modelling the interaction between this relation and CEOs' tenure, thereby examining how the relation evolves with CEOs' tenure.

By examining a sample of Compustat ExecuComp firms from 1993 to 2007 for which the GAI has been constructed by Custódio et al. (2013), we obtain two major observations. We first find CEOs' general managerial skills to have a significantly positive effect on firm risk for CEOs in the early years of their tenure. Second, we find that the positive relation between firm risk and the GAI weakens as the CEO's tenure increases. The weakening effect of tenure on the GAI–firm risk relation is statistically highly significant and economically strong. With CEO and firm characteristics and industry-by-year fixed effects controlled, the effects of the GAI on firm risk are significantly positive for CEOs in the early 4 years of tenure (tenure years 0–3) and become insignificant in the next 2 years (tenure years 4 and 5) and significantly negative thereafter (tenure years greater than 5).

In support of the above main findings, CEOs' general managerial skills have similar effects on the riskiness of corporate policies. For example, in the early stage of tenure, generalists are more aggressive than specialists in terms of research and development (R&D) policy and patent applications, but the difference attenuates with tenure. In addition, relative to specialists, generalists tend to use fewer fixed costs versus variable costs and implement a higher degree of diversification strategy the longer they manage their firms.

Our second main finding, tenure's weakening of the relation between the GAI and firm risk, highlights the important role of CEO tenure in the general to firm-specific managerial skills transformation process. Intuitively, we expect such a transformation to be more relevant for new CEOs who start as outsiders than for those who start as insiders, since insiders' human capital and business experiences could have long been confined to the same firm, thus becoming largely firm specific. Indeed, in our sample, insider CEOs had an average tenure of 18 years with their firm before assuming the CEO position. Hence, we expect a notable difference in the effect of general managerial skills between inside and outside CEOs. To investigate this difference, we perform a subsample analysis of outside and inside CEOs in which outside CEOs are defined as those with preappointment tenure within the median of the total sample (i.e., 4 years). Consistent with our conjecture, we find that our main findings of the initial positive effect of CEOs' general managerial skills on firms' stock return volatility and the pattern of tenure weakening the effect are mainly driven by outside CEOs.

By conducting robustness checks, we rule out potential alternative explanations for our findings, such as the change in generalists' managerial preferences with tenure or the performance-driven risk-taking behaviour of generalist CEOs. We also rule out the possibility that our findings reflect the pattern of outside CEOs' risk-taking behaviour or generalist CEOs' risk-taking profile with age. Using the methods of propensity score matching (PSM), firm or CEO fixed effects and instrumental variables (IVs), we show that our main results are robust to endogeneity concerns.

Our paper is related to the literature studying the effects of CEOs' general managerial skills. Although the shift in the relative importance of general managerial skills was first emphasized by Murphy and Zábojník (2004, 2007) as a potentially important determinant of the upward trend in CEO compensation over the three decades before the early 2000s, this CEO characteristic was not extensively investigated until Custódio et al. (2013) developed the GAI and made the data publicly available. Until now, researchers have examined the effects of CEOs' general managerial skills on CEO compensation, corporate strategic policies and firm-level outcomes. Contributing to this literature, our study addresses an important dimension of general managerial skills that have not yet been noted, that is, CEO tenure. This dimension raises serious issues regarding the characterization of generalist CEOs. When managerial skills undergo the transformation from general to firm specific with managerial tenure, generalist CEOs staying with the same firm are bound to become specialists at a certain stage. An important implication is that any effects of general managerial skills are likely to be a decreasing function of managerial tenure and are expected to disappear with long-tenured CEOs.

Our paper is also related to the literature studying the determinants of CEOs' risk-taking behaviour. Existing research has identified a vast number of determinants of CEOs' risk-taking behaviour, such as the compensation incentives provided to CEOs (e.g., Cassell et al., 2012; Coles et al., 2006; Do et al., 2022) and CEO characteristics and experience (e.g., Bernile et al., 2017; Cain & McKeon, 2016; Faccio et al., 2016; Kamiya et al., 2019; Serfling, 2014). Our paper adds to this stream of literature by showing that CEO managerial skill generality is a

managerial trait that also plays a role in affecting CEOs' risk-taking behaviour, although its role depends on the stage of CEO tenure.

More broadly, our paper is related to the literature investigating the effects of CEO characteristics and experience on corporate policies. These studies include those of Benmelech and Frydman (2015), Bertrand and Schoar (2003), Custódio and Metzger (2014), Islam and Zein (2020), Malmendier and Tate (2005, 2008), Malmendier et al. (2011) and Schoar and Zuo (2017), among many others.

The remainder of this paper is organized as follows. Section 2 describes the sample and data. Section 3 reports our baseline results on the relation between CEOs' general managerial skills and firm risk and the role of CEO tenure in weakening this relation. Section 3 also examines the mechanisms of our baseline results. Section 4 addresses endogeneity concerns. Section 5 investigates alternative explanations for our results. Section 6 provides the conclusion.

2 | SAMPLE AND DATA

2.1 | Sample

As with Custódio et al. (2013), our sample consists of Compustat ExecuComp firms from 1993 to 2007 whose CEOs' prior professional experience can be identified in the BoardEx database. The sample consists of 21,909 firm-year observations, with 2651 unique firms and 4451 different CEOs. The firm financial variables and stock return data are retrieved from Compustat and the Center for Research in Security Prices (CRSP), respectively. The definitions of the variables are provided in Table A1 in Appendix A.

2.2 | Measures of CEOs' general managerial skills

Our variable of interest, a proxy for CEOs' general managerial skills, is the GAI, constructed by Custódio et al. (2013).¹ The GAI variable, *GAI*, is measured as the first factor from applying the principal component factor to the following five proxies of CEOs' general managerial ability: (1) the past number of positions (*X1*), (2) the past number of firms (*X2*), (3) the past number of industries in which the CEO has worked (*X3*), (4) whether the CEO held a CEO position at a different company (*X4*) and (5) whether the CEO worked for a conglomerate (*X5*). The formula for the GAI is as follows:

$$GAI_{i,t} = 0.268X1_{i,t} + 0.312X2_{i,t} + 0.309X3_{i,t} + 0.218X4_{i,t} + 0.153X5_{i,t}. \quad (1)$$

As discussed by Custódio et al. (2013), all five variables have positive loadings and are, thus, positively related to the GAI. Therefore, a greater value of the GAI indicates a higher level of general managerial ability. Then, the index is standardized to have a mean of zero and a standard deviation of one. Table 1 reports the summary statistics of selected CEO and firm characteristic variables, where all continuous variables are winsorized at their first and 99th

¹We thank Custódio, Ferreira and Matos for making these data publicly available. Their sample and data are available at <https://www.jfinec.com/data-and-code>.

TABLE 1 Summary statistics

This table presents the summary statistics of the CEO and firm characteristics. The sample consists of Compustat ExecuComp firms from 1993 to 2007 whose CEOs' prior professional experience can be identified in the BoardEx database. The definitions of all variables are provided in Table A1 in Appendix A. The continuous variables are winsorized at their first and 99th percentiles.

	Mean	Median	Std dev	Min	Max	Obs
Panel A: CEO characteristics						
<i>General Ability Index (GAI)</i>	−0.03	−0.18	0.92	−1.42	2.97	21,583
<i>Generalist Dummy</i>	0.50	0.00	0.50	0.00	1.00	21,909
<i>Tenure</i>	6.60	5.00	6.43	0.00	34.00	19,905
<i>CEO Age</i>	55.43	56.00	6.81	39.00	75.00	20,513
<i>CEO Delta (\$ thousands)</i>	689.86	232.68	1461.00	3.23	15,116.38	19,219
<i>CEO Vega (\$ thousands)</i>	112.73	46.62	176.66	0.00	1316.25	20,159
<i>Female Dummy</i>	0.01	0.00	0.12	0.00	1.00	21,909
Panel B: Firm characteristics						
<i>Total Risk</i>	0.40	0.36	0.19	0.14	1.18	19,013
<i>Idiosyncratic Risk</i>	0.36	0.31	0.18	0.12	1.06	19,022
<i>Operating Assets Volatility</i>	0.28	0.23	0.21	0.02	1.26	18,958
<i>R&D Intensity</i>	0.02	0.00	0.05	0.00	0.27	21,663
<i>Patents</i>	20.13	0.00	62.24	0.00	627.00	15,360
<i>Book Leverage</i>	0.22	0.21	0.17	0.00	0.83	21,579
<i>Operating Leverage</i>	1.94	1.48	3.72	−23.20	29.54	15,166
<i>Segments</i>	2.74	2.00	1.95	1.00	9.00	19,057
<i>Total Assets (\$ millions)</i>	7853	1655	19,729	63	217,123	21,445
<i>Tobin's Q</i>	1.94	1.52	1.21	0.80	8.96	21,394
<i>Asset Tangibility</i>	0.28	0.22	0.23	0.00	0.88	20,936
<i>ROA</i>	0.09	0.09	0.08	−0.30	0.36	21,103
<i>Sales Growth</i>	0.13	0.09	0.23	−0.47	1.58	21,408
<i>Firm Age</i>	22.36	17.00	18.13	0.00	78.00	19,735

Abbreviations: CEO, chief executive officer; Max, maximum; Min, minimum; Obs, Observations; Std dev, standard deviation.

percentiles. The mean, median and standard deviations of *GAI* are −0.03, −0.18 and 0.92, respectively.

Consistent with Custódio et al. (2013) and others, we also define a *Generalist Dummy*, a dummy variable that assumes the value of one if the *GAI* is greater than the yearly median, and zero otherwise. The CEOs with a *GAI* no greater than the yearly median are classified as specialists. The variable *Generalist Dummy* is used in our robustness checks as an alternative measure of CEOs' general managerial skills and in the PSM method that we discuss in Section 4.

2.3 | Measures of firm risk and risk-related corporate policies

We define a series of variables as proxies for firm risk and risk-related corporate policies. For the firm risk measures, we define *Total Risk*, *Idiosyncratic Risk* and *Operating Assets Volatility*. Following Serfling (2014), *Total Risk* is measured as the annualized standard deviation of the daily stock returns of the fiscal year, and *Idiosyncratic Risk* is measured as the annualized standard deviation of the residuals from the regressions of the daily stock return on Fama–French three factors estimated over the fiscal year.² The mean (median) value of *Total Risk* is 0.40 (0.36), and the mean (median) value of *Idiosyncratic Risk* is 0.36 (0.31), values that are comparable to but lower than those reported by Serfling (2014), whose sample largely overlaps with ours but covers a longer period, including the financial crisis period from 2008 to 2009. Similar to Gormley and Matsa (2016), we define *Operating Assets Volatility* as $Total\ Risk \times [E/(V - C)]$, where E is a firm's market value, and $V - C$ is the firm's enterprise value. The mean and median of *Operating Assets Volatility* are 0.28 and 0.23, respectively.

We follow the literature (e.g., Cain & McKeon, 2016; Kini & Williams, 2012; Serfling, 2014) to identify firm policies that have a direct impact on firm risk. We first define *R&D Intensity*, which is calculated as the ratio of a firm's R&D expenses to its total assets, with missing values of R&D expenses set to zero. Furthermore, we define *Patents* as a measure of the productivity of R&D expenses. In particular, to consider the time lag between the application date and the grant date of patents (on average, 2 years), we follow the innovation literature (e.g., Custódio et al., 2019; Hall et al., 1986; Islam & Zein, 2020) and measure *Patents* as the number of patents filed by a firm in a fiscal year. The patent data are obtained from the extended data set of Kogan et al. (2017; henceforth KPSS), which provides patent-level data, including the unique identification, estimated market values and the numbers of citations, of patents issued by the US Patent and Trademark Office from 1926 to 2020.³ Given that our sample period ends in 2007, the typical time truncation concern is not present (i.e., filed patents have not yet been granted and thus do not appear in the data set). The mean (median) value of *R&D Intensity* is 0.02 (0.00). The mean (median) value of *Patents* is 20.13 (0.00).

We define *Book Leverage* as the ratio of a firm's long-term debt and debt in current liabilities to total assets. The mean and median of *Book Leverage* are 0.22 and 0.21, respectively, which are very close to those reported in other studies.

Next, we closely follow Serfling (2014) to construct *Operating Leverage*, which is defined as the elasticity of a firm's operating income with respect to its sales and is a measure of the relative proportion of fixed costs versus variable costs incurred in a firm. We use Compustat quarterly data over a 3-year window of the years t to $t + 2$ and require that the CEO in year t hold the CEO position for at least two more years (until $t + 2$) and that a firm-year have nonmissing data for at least eight quarters. For each firm-year, we run the following model over years $[t, t + 2]$ with the quarterly data:

$$OIADPQ_{i,t,q} = \alpha_{i,t} + \pi_{i,t} SALE_{i,t,q} + \varepsilon_{i,t,q}, \quad (2)$$

²The Fama–French three factors are obtained from Kenneth R. French's website (https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html).

³The data are available at <https://github.com/KPSS2017/Technological-Innovation-Resource-Allocation-and-Growth-Extended-Data>.

where i , t and q denote the firm, year and quarter, respectively; $OIADPQ$ denotes the quarterly operating income after depreciation; and $SALE$ denotes the quarterly total revenue. The estimated coefficient ($\widehat{\pi}_{i,t}$) on $SALE$ represents the dollar change in the operating income in response to a dollar increase in sales. Therefore, *Operating Leverage* is calculated as $\widehat{\pi}_{i,t} / (\overline{SALE}_{i,t,q} / \overline{OIADPQ}_{i,t,q})$, where $\overline{SALE}_{i,t,q}$ and $\overline{OIADPQ}_{i,t,q}$ denote the 3-year average of quarterly sales and operating income over the years $[t, t+2]$ for firm i in year t , respectively. The mean (median) value of *Operating Leverage* is 1.94 (1.48), which is comparable to the statistics reported by Serfling (2014).

Finally, we use the natural logarithm of the number of business segments ($Ln(Segments)$) to measure the degree of a firm's diversification. The data on the firms' business segments are obtained from the Compustat Historical Segments database. The mean (median) number of business segments is 2.74 (2.00).

2.4 | Other explanatory variables

We follow the literature (e.g., Cain & McKeon, 2016; Faccio et al., 2016; Kini & Williams, 2012; Serfling, 2014) and choose several CEO and firm characteristic variables as control variables. CEO wealth-related incentives for risk taking are measured as the natural logarithm of one plus delta (the dollar change in the portfolio value for a 1% change in the stock price, or *CEO Delta*) and the natural logarithm of one plus vega (the dollar change in the portfolio value for a 0.01 change in the annualized standard deviation of stock returns, or *CEO Vega*), where delta and vega are calculated by Coles et al. (2006).⁴ The other CEO characteristic variables, including *Tenure*, *CEO Age* and *Female Dummy*, are obtained from the Compustat ExecuComp database. The average CEO age and tenure are 55.4 and 6.6 years, respectively. Only approximately 1% of the CEOs is female.

The firm characteristic variables include *Ln(Assets)*, *Tobin's Q*, *Asset Tangibility*, *ROA*, *Sales Growth* and *Ln(Firm Age)*, where *Ln(Assets)* is the natural logarithm of a firm's total assets, with a mean (median) value of total assets of 7853 million USD (1655 million USD); *Ln(Firm Age)* is the natural logarithm of the firm's age plus one, with a mean (median) value of firm age of 22.4 (17.0); *Tobin's Q* is the market value of a firm's total assets divided by the book value of its total assets; *Asset Tangibility* is net property, plant and equipment divided by total assets; *ROA* is earnings before interest and taxes divided by total assets; and *Sales Growth* is measured as $(SALE_t / SALE_{t-1}) - 1$. The mean values of *Tobin's Q*, *Asset Tangibility*, *ROA* and *Sales Growth* are 1.94, 0.28, 0.09 and 0.13, respectively.

2.5 | CEOs' general managerial skills and CEO and firm characteristics

To obtain a general impression of the differences between generalists and specialists and the firms they run, we conduct a comparison of the CEO and firm characteristic variables between

⁴On the basis of the definitions of delta and vega of Coles et al. (2006) and Core and Guay (2002), Coles, Daniel and Naveen provide detailed instructions regarding how to construct these variables in their article entitled "Calculation of Compensation Incentives and Firm-related Wealth Using ExecuComp: Data, Program and Explanation". They also make the data publicly available at Naveen's website (<https://sites.temple.edu/lnaveen/data/>). We thank these authors for providing these data.

the generalist and specialist samples (see Table 2). We observe that all the variables are statistically different between the two samples. Regarding personal characteristics, generalists tend to be older and have a shorter tenure with a firm. Moreover, generalists are associated with a significantly higher delta than specialists, which is consistent with Liu et al. (2020), and a significantly higher vega. The most prominent difference in firm characteristics is that firms

TABLE 2 Statistics of generalist–specialist differences

This table presents the statistics of the CEO and firm characteristics of the generalist and specialist sample and the statistics of the generalist–specialist differences. CEOs with *GAI* greater than the yearly median are identified as generalists; otherwise, the CEOs are identified as specialists. Panels A and B present the statistics of the CEO and firm characteristics, respectively. The definitions of all variables are provided in Table A1 in Appendix A. The continuous variables are winsorized at their first and 99th percentiles. ***, ** and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

	Generalist		Specialist		Generalist–specialist difference	
	Mean	Median	Mean	Median	Mean	Median
Panel A: CEO characteristics						
<i>GAI</i>	0.76	0.54	−0.77	−0.74	1.53***	1.28***
<i>Tenure</i>	5.78	4.00	7.50	5.00	−1.72***	−1.00***
<i>CEO Age</i>	56.30	57.00	54.59	55.00	1.71***	2.00***
<i>CEO Delta (\$ thousands)</i>	728.64	251.24	656.49	215.97	72.15***	35.27***
<i>CEO Vega (\$ thousands)</i>	150.97	64.95	77.28	34.43	73.69***	30.52***
<i>Female Dummy</i>	0.02	0.00	0.01	0.00	0.01***	0.00***
Panel B: Firm characteristics						
<i>Total Risk</i>	0.39	0.34	0.41	0.37	−0.02***	−0.03***
<i>Idiosyncratic Risk</i>	0.35	0.30	0.37	0.33	−0.02***	−0.03***
<i>Operating Assets Volatility</i>	0.27	0.21	0.30	0.24	−0.03***	−0.03***
<i>R&D Intensity</i>	0.03	0.00	0.02	0.00	0.00***	0.00***
<i>Patents</i>	29.97	1.00	10.79	0.00	19.18***	1.00***
<i>Book Leverage</i>	0.24	0.23	0.21	0.19	0.03***	0.04***
<i>Operating Leverage</i>	1.81	1.46	2.06	1.50	−0.25***	−0.04***
<i>Segments</i>	3.05	3.00	2.47	2.00	0.58***	1.00***
<i>Total Assets (\$ millions)</i>	11,277	2433	4877	1117	6399***	1316***
<i>Tobin's Q</i>	1.89	1.51	1.99	1.53	−0.09***	−0.02*
<i>Asset Tangibility</i>	0.29	0.23	0.28	0.21	0.01***	0.02***
<i>ROA</i>	0.09	0.08	0.09	0.09	−0.01***	−0.01***
<i>Sales Growth</i>	0.12	0.08	0.15	0.10	−0.02***	−0.02***
<i>Firm Age</i>	25.11	20.00	19.92	16.00	5.19***	4.00***

Abbreviations: CEO, chief executive officer; GAI, General Ability Index.

run by generalists are significantly larger. In terms of total assets, firms run by generalists are more than twice as large as those run by specialists. In addition, compared with firms run by specialists, firms run by generalists are older and, consistent with Custódio et al. (2019), have more patents.

3 | EMPIRICAL RESULTS

3.1 | CEOs' general managerial skills, tenure and firm risk

In this subsection, we examine the effect of CEOs' general managerial skills on firm risk and how the effect evolves with CEO tenure. As discussed earlier, newly appointed generalist CEOs can be more risk loving than their newly appointed specialist counterparts because the former's personal wealth is less contingent on the performance of the current firm due to their richer outside job opportunities. However, with the passage of time, generalists' managerial skills gradually become more firm specific and less applicable to other firms, rendering them more risk averse and deterring them from taking risks. Therefore, we predict that CEOs' general managerial skills have a positive effect on firm risk in the early stage of CEO tenure, with a pattern of the effect that weakens with tenure. Table 3 reports the results of the ordinary least squares (OLS) regression analysis, with *Total Risk* being the dependent variable. In Columns (1)–(3), we include industry and year fixed effects separately, where the industry fixed effects are determined based on a firm's two-digit Standard Industrial Classification (SIC) code.⁵ To account for the possibility that the industry effects on firm–CEO matching or firm risk are time varying, in Columns (4)–(6) we include industry $\times$ year fixed effects. All continuous variables are winsorized at their first and 99th percentiles.

We first examine the overall effect of CEOs' general managerial skills on firm risk. Previously, we only predict a positive effect of the GAI on firm risk in the early stage of CEO tenure, but the effect of the GAI on firm risk over the entire course of a CEO's tenure is less clear. In Columns (1) and (4) of Table 3, we regress *Total Risk* on *GAI*, $\ln(\text{Tenure})$, the control variables and the fixed effects. The control variables include $\ln(\text{CEO Age})$, $\ln(\text{CEO Delta})$, $\ln(\text{CEO Vega})$, *Female Dummy*, $\ln(\text{Assets})$, *Book Leverage*, *Tobin's Q*, *Asset Tangibility*, *R&D Intensity*, *ROA*, *Sales Growth* and $\ln(\text{Firm Age})$, which are henceforth referred to as the baseline set of controls. The estimated coefficients on *GAI* in Columns (1) and (4) are positive and statistically insignificant, indicating that the overall effect of CEOs' general managerial skills on firm total risk is positive but weak in economic magnitude. On the basis of the results in Column (4), the estimated coefficient suggests that as *GAI* increases by one standard deviation (0.92), *Total Risk*, on average, increases by 0.00184 (0.002×0.92), accounting for only 0.46% ($0.00184/0.40$) of the mean value of *Total Risk*. This effect is much weaker than the effects on firm risk of many other CEO characteristics.

The effects of the control variables on *Total Risk* are largely consistent with those documented in the literature (e.g., Cain & McKeon, 2016; Kini & Williams, 2012; Serfling, 2014). Specifically, CEO age ($\ln(\text{CEO Age})$) and vega ($\ln(\text{CEO Vega})$) are negatively associated with *Total Risk*. Among firm characteristics, firm size ($\ln(\text{Assets})$), age ($\ln(\text{Firm Age})$), *Asset*

⁵As a robustness check, throughout the paper, we also classify the industry groupings based on firms' three SIC codes and run similar regressions. The alternative definition of industry fixed effects does not qualitatively change our results.

TABLE 3 CEOs' general managerial skills, tenure and firms' total risk

This table presents the regression results of the relation between CEOs' general managerial skills and firm risk and the evolution of the relation with CEOs' tenure. The sample consists of Compustat ExecuComp firms from 1993 to 2007 whose CEOs' prior professional experience can be identified in the BoardEx database. The dependent variable is *Total Risk*, which is defined as the annualized daily stock return in a fiscal year. The key independent variable is *GAI*, which is an index constructed by Custódio et al. (2013) as a measure of a CEO's general managerial ability. The mediating variable $\ln(\textit{Tenure})$ is the natural logarithm of *Tenure* plus one. The industry fixed effects are determined based on the firm's two-digit SIC code. The definitions of the other variables are provided in Table A1 in Appendix A. All continuous variables are winsorized at their first and 99th percentiles. Standard errors are clustered by firm, and heteroskedasticity robust *p* values are provided in parentheses. ***, ** and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Constant</i>	0.891*** (0.000)	0.975*** (0.000)	0.914*** (0.000)	0.924*** (0.000)	1.027*** (0.000)	1.038*** (0.000)
<i>GAI</i>	0.003 (0.185)	0.016*** (0.000)	0.006 (0.202)	0.002 (0.268)	0.018*** (0.000)	0.009* (0.054)
$GAI \times \ln(\textit{Tenure})$		-0.008*** (0.000)			-0.009*** (0.000)	
$GAI \times \textit{Dummy}(\textit{Tenure} = 1)$			0.009* (0.059)			0.007 (0.125)
$GAI \times \textit{Dummy}(\textit{Tenure} = 2)$			0.003 (0.612)			0.001 (0.851)
$GAI \times \textit{Dummy}(\textit{Tenure} = 3)$			0.004 (0.502)			-0.000 (0.989)
$GAI \times \textit{Dummy}(\textit{Tenure} = 4)$			0.002 (0.706)			-0.003 (0.598)
$GAI \times \textit{Dummy}(\textit{Tenure} = 5)$			-0.000 (0.998)			-0.005 (0.366)
$GAI \times \textit{Dummy}(\textit{Tenure} > 5)$			-0.010** (0.049)			-0.015*** (0.005)
$\ln(\textit{Tenure})$	0.006** (0.012)	0.005** (0.023)	0.011* (0.075)	0.006*** (0.008)	0.006** (0.016)	0.009 (0.162)
$\ln(\textit{CEO Age})$	-0.086*** (0.000)	-0.088*** (0.000)	-0.091*** (0.000)	-0.083*** (0.000)	-0.086*** (0.000)	-0.087*** (0.000)
$\ln(\textit{CEO Delta})$	-0.001 (0.512)	-0.002 (0.410)	-0.002 (0.305)	-0.003 (0.209)	-0.003 (0.145)	-0.003 (0.111)
$\ln(\textit{CEO Vega})$	-0.007*** (0.000)	-0.007*** (0.000)	-0.006*** (0.000)	-0.005*** (0.000)	-0.005*** (0.001)	-0.005*** (0.001)

TABLE 3 (Continued)

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Female Dummy</i>	0.002 (0.928)	0.002 (0.927)	0.001 (0.941)	−0.004 (0.838)	−0.003 (0.844)	−0.004 (0.833)
<i>Ln(Assets)</i>	−0.024*** (0.000)	−0.024*** (0.000)	−0.023*** (0.000)	−0.024*** (0.000)	−0.024*** (0.000)	−0.024*** (0.000)
<i>Book Leverage</i>	0.047*** (0.001)	0.048*** (0.001)	0.047*** (0.001)	0.049*** (0.001)	0.050*** (0.000)	0.049*** (0.001)
<i>Tobin's Q</i>	0.017*** (0.000)	0.017*** (0.000)	0.017*** (0.000)	0.016*** (0.000)	0.016*** (0.000)	0.016*** (0.000)
<i>Asset Tangibility</i>	−0.042*** (0.005)	−0.042*** (0.004)	−0.042*** (0.005)	−0.041*** (0.009)	−0.041*** (0.008)	−0.041*** (0.008)
<i>R&D Intensity</i>	0.725*** (0.000)	0.716*** (0.000)	0.714*** (0.000)	0.713*** (0.000)	0.704*** (0.000)	0.702*** (0.000)
<i>ROA</i>	−0.590*** (0.000)	−0.585*** (0.000)	−0.583*** (0.000)	−0.602*** (0.000)	−0.596*** (0.000)	−0.595*** (0.000)
<i>Sales Growth</i>	0.053*** (0.000)	0.054*** (0.000)	0.054*** (0.000)	0.059*** (0.000)	0.060*** (0.000)	0.060*** (0.000)
<i>Ln(Firm Age)</i>	−0.030*** (0.000)	−0.030*** (0.000)	−0.030*** (0.000)	−0.029*** (0.000)	−0.028*** (0.000)	−0.029*** (0.000)
<i>Tenure Group Dummies</i>	No	No	Yes	No	No	Yes
<i>Industry Fixed Effects</i>	Yes	Yes	Yes	No	No	No
<i>Year Fixed Effects</i>	Yes	Yes	Yes	No	No	No
<i>Industry × Year Fixed Effects</i>	No	No	No	Yes	Yes	Yes
Observations	12,797	12,797	12,797	12,797	12,797	12,797
Adjusted R ²	0.594	0.595	0.595	0.625	0.627	0.627
Marginal effects of <i>GAI</i>						
<i>Tenure = 0</i>			0.006 (0.202)			0.009* (0.054)
<i>Tenure = 1</i>			0.015*** (0.000)			0.016*** (0.000)
<i>Tenure = 2</i>			0.008** (0.018)			0.010*** (0.006)
<i>Tenure = 3</i>			0.010** (0.014)			0.009** (0.023)
<i>Tenure = 4</i>			0.008** (0.038)			0.006 (0.119)

(Continues)

TABLE 3 (Continued)

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Tenure</i> = 5			0.006 (0.192)			0.003 (0.422)
<i>Tenure</i> > 5			−0.004 (0.121)			−0.006** (0.042)

Abbreviations: CEO, chief executive officer; GAI, General Ability Index.

Tangibility and operating performance (*ROA*) have significantly negative effects on *Total Risk*, whereas *Book Leverage*, *Tobin's Q*, *R&D Intensity* and *Sales Growth* have significantly positive effects on *Total Risk*.

To examine the evolution of the relation between *GAI* and *Total Risk* with CEO tenure, in Columns (2) and (5) of Table 3, we further include the interaction term between *GAI* and *Ln (Tenure)*. The results in both columns are consistent with our conjecture: The coefficients on *GAI* are significantly positive and the coefficients on *GAI* × *Ln(Tenure)* are significantly negative, indicating that the effect of CEOs' general managerial skills on firm total risk is significantly positive in the earlier years but significantly diminishes with CEO tenure.⁶ These results are consistent with our conjecture. With more outside job opportunities, generalists are more risk loving than specialists in the early stage of their tenure. However, the longer they manage their firms, generalists' managerial skills become increasingly firm specific, deterring them from job switching and from taking extra risks in the current firm. Another testable prediction from this conjecture is that, because generalists' managerial skills are expected to gradually become vested in the firm, the effect of tenure weakening the relation between firm risk and CEOs' general managerial skills should be monotonic or at least quasimonotonic.

To further illuminate the effect of CEO tenure on the relation between CEOs' general managerial skills and firm total risk and particularly determine whether the weakening effect of *GAI* on *Total Risk* is monotonic (or quasimonotonic) throughout a CEO's tenure, we use dummies for tenure years. Specifically, for each tenure year from one to five, we define a dummy variable (*Dummy(Tenure = t)*) to denote that year (*Dummy(Tenure = t)* equals one if *Tenure = t*, and zero otherwise). In addition, we define *Dummy(Tenure > 5)* as a binary variable that equals one if *Tenure* is greater than 5, and zero otherwise. Therefore, CEOs with tenure year of zero are treated as the default group. Then, we interact *GAI* with each tenure group dummy. Columns (3) and (6) of Table 3 report the results of the OLS regressions in which the coefficient on an interaction term between *GAI* and a tenure group dummy represents the difference in the effect of *GAI* on *Total Risk* between that specific tenure group and the default group. To facilitate interpreting the results, we then calculate the marginal effects on *Total Risk* of *GAI* for each tenure group and report them below the regression results. The marginal effects show a quasimonotonic trend as follows: Except for *Tenure* equal to zero, which represents CEOs who have assumed the current position for less than a full year, the marginal effects of *GAI* on *Total Risk* generally decrease with CEO tenure. According to Column (6), the effects of *GAI* on *Total Risk* are significantly positive in the early 4 years of the CEO tenure (tenure years

⁶As a robustness check, we use *Generalist Dummy* as an alternative measure of CEOs' general managerial skills and obtain qualitatively similar results.

0–3) and become insignificant in the next 2 years (tenure years 4 and 5) and significantly negative thereafter (tenure years greater than 5). This quasimonotonic weakening effect of *GAI* on *Total Risk* is consistent with the notion that the general to firm-specific managerial skills transformation is a gradual and continuing process.

The hypothesis of the general to firm-specific managerial skills transformation implies that the generality of CEOs' managerial skills decreases with CEOs' tenure. Therefore, as with Betzer et al. (2020), we examine the determinants of the *GAI* and pay special attention to its relation with CEOs' tenure. In particular, we regress *GAI* on $\ln(\text{Tenure})$, the baseline control variables and the fixed effects (separate industry and year fixed effects or industry $\times$ year fixed effects). Focusing on small samples of turnover executives, Betzer et al. (2020) obtain mixed results for the relation between CEOs' tenure and the *GAI*.⁷ However, using a much larger sample consisting of CEOs with available *GAI*, we show in the untabulated results a significantly negative association between the *GAI* and CEOs' tenure, a finding that is consistent with our conjectured hypothesis of general to firm-specific managerial skills transformation.

As a robustness check for our findings regarding the effect of the *GAI* on *Total Risk*, we use alternative measures of firm risk. In particular, we use *Idiosyncratic Risk* and *Operating Assets Volatility* as dependent variables and run the same regressions as specifications (2) and (5) in Table 3. The results reported in Table 4 show a similar pattern: The *GAI* has significantly positive effects on *Idiosyncratic Risk* and *Operating Assets Volatility* in the early years of CEOs' tenure, and the effects decrease significantly with CEOs' tenure.

Collectively, consistent with the conjecture of general to firm-specific managerial skills transformation, we find that CEOs' general managerial skills have a significantly positive effect on firm risk in the early stage of a CEO's tenure and that the effect significantly weakens with tenure. The weakening effect of tenure exhibits a quasimonotonic pattern, and the findings are robust to alternative measures of firm risk.

3.2 | CEOs' general managerial skills, tenure and corporate policies

In Section 3.1, we present the effect of tenure weakening the relation between CEOs' general managerial skills and firm risk. In this subsection, we examine the channel through which firm risk is impacted. In particular, we investigate whether the effects of CEOs' general managerial skills on the riskiness of corporate policies also exhibit a pattern that weakens with tenure. We follow the literature and consider policies relevant to a firm's risk, such as R&D intensity, patenting, book leverage, operating leverage and firm diversification, and use each policy as a dependent variable that is regressed on *GAI*, $\text{GAI} \times \ln(\text{Tenure})$, $\ln(\text{Tenure})$, the control variables and the fixed effects. The OLS regression results of this analysis are reported in Table 5. Panel A reports the base regressions, which include industry and year fixed effects separately, and Panel B reports the results of the regressions that include industry $\times$ year fixed effects. The control variables are from the baseline set, with minor adjustments based on the dependent variable.

⁷Focusing on the sample of sudden executive deaths, Betzer et al. (2020) find an insignificant relation between CEOs' tenure and the *GAI*. Whereas, with the sample of nonsudden exogenous CEO turnovers or the pooled sample of all CEO turnovers, Betzer et al. (2020) show that the relation between CEOs' tenure and the *GAI* is significantly negative.

TABLE 4 CEOs' general managerial skills, tenure and alternative risk measures

This table presents the regression results of the relation between CEOs' general managerial skills and firms' alternative risk measures and the evolution of the relation with CEOs' tenure. The sample consists of Compustat ExecuComp firms from 1993 to 2007 whose CEOs' prior professional experience can be identified in the BoardEx database. The dependent variables are *Idiosyncratic Risk* in Columns (1) and (2) and *Operating Assets Volatility* in Columns (3) and (4). The key independent variable is *GAI*, which is an index constructed by Custódio et al. (2013) as a measure of a CEO's general managerial ability. The mediating variable $\ln(\text{Tenure})$ is the natural logarithm of *Tenure* plus one. The control variables include $\ln(\text{CEO Age})$, $\ln(\text{CEO Delta})$, $\ln(\text{CEO Vega})$, *Female Dummy*, $\ln(\text{Assets})$, *Book Leverage*, *Tobin's Q*, *Asset Tangibility*, *R&D Intensity*, *ROA*, *Sales Growth* and $\ln(\text{Firm Age})$. The industry fixed effects are determined based on the firm's two-digit SIC code. The definitions of the other variables are provided in Table A1 in Appendix A. All continuous variables are winsorized at their first and 99th percentiles. Standard errors are clustered by firm, and heteroskedasticity robust *p* values are provided in parentheses. ***, ** and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

	<i>Idiosyncratic Risk</i>		<i>Operating Assets Volatility</i>	
	(1)	(2)	(3)	(4)
<i>Constant</i>	0.970*** (0.000)	0.993*** (0.000)	0.873*** (0.000)	0.923*** (0.000)
<i>GAI</i>	0.017*** (0.000)	0.018*** (0.000)	0.009** (0.018)	0.011*** (0.003)
$GAI \times \ln(\text{Tenure})$	-0.008*** (0.000)	-0.009*** (0.000)	-0.004** (0.025)	-0.006*** (0.003)
$\ln(\text{Tenure})$	0.004** (0.043)	0.005** (0.032)	-0.002 (0.495)	-0.001 (0.568)
<i>Control Variables</i>	Yes	Yes	Yes	Yes
<i>Industry Fixed Effects</i>	Yes	No	Yes	No
<i>Year Fixed Effects</i>	Yes	No	Yes	No
<i>Industry $\times$ Year Fixed Effects</i>	No	Yes	No	Yes
Observations	12,809	12,809	12,841	12,841
Adjusted R^2	0.604	0.628	0.681	0.701

Abbreviations: CEO, chief executive officer; GAI, General Ability Index.

We first investigate a firm's R&D policy and use *R&D Intensity* as the dependent variable in Column (1) of Table 5. Due to the risky nature of R&D expenses, a higher value of *R&D Intensity* indicates a firm's riskier policies. Naturally, *R&D Intensity* is removed from the control variables. The results in Column (1) are very consistent with our main findings: The coefficients on *GAI* are significantly positive and the coefficients on $GAI \times \ln(\text{Tenure})$ are significantly negative, indicating that, relative to specialists, generalist CEOs implement more aggressive R&D policies in the earlier stage than in the later stage of their tenure.

Furthermore, regarding R&D policies, we consider the productivity of R&D expenses and use $\ln(\text{Patent})$ as the dependent variable in Column (2) of Table 5. While constructing the sample, we follow Custódio et al. (2019) and exclude financial firms (SIC codes 6000–6999),

TABLE 5 CEOs' general managerial skills, tenure and corporate policies

This table presents the regression results of the relation between CEOs' general managerial skills and the riskiness of firms' policies and the evolution of the relation with CEOs' tenure. The sample consists of Compustat ExecuComp firms from 1993 to 2007 whose CEOs' prior professional experience can be identified in the BoardEx database. Column (2) excludes financial firms (SIC codes 6000–6999), transportation and utilities firms (SIC codes 4000–4999) and firms that operate in four-digit SIC industries that have never filed any patents according to the extended KPSS data set. Panel A reports the results of the regressions, including industry and year fixed effects. Panel B reports the results of the regressions, including industry $\times$ year fixed effects. The dependent variables in Columns (1)–(5) are *R&D Intensity*, *Ln(Patents)* (the natural logarithm of the number of patents plus one), *Book Leverage*, *Operating Leverage* and *Ln(Segments)*, respectively. The key independent variable is *GAI*, which is an index constructed by Custódio et al. (2013) as a measure of a CEO's general managerial ability. The mediating variable *Ln(Tenure)* is the natural logarithm of *Tenure* plus one. The control variables include *Ln(CEO Age)*, *Ln(CEO Delta)*, *Ln(CEO Vega)*, *Female Dummy*, *Ln(Assets)*, *Book Leverage*, *Tobin's Q*, *Asset Tangibility*, *R&D Intensity*, *ROA*, *Sales Growth* and *Ln(Firm Age)* after minor adjustments. Columns (1) and (3) omit *R&D Intensity* and *Book Leverage*, respectively. Columns (2) and (4) add *CAPEX* and *Labour Stock*, respectively, as additional controls. Column (5) adds *Free Cash Flow* and *CAPEX* as additional control variables. The industry fixed effects are determined based on the firm's two-digit SIC code. The definitions of the variables are provided in Table A1 in Appendix A. All continuous variables are winsorized at their first and 99th percentiles. Standard errors are clustered by firm, and heteroskedasticity robust *p* values are provided in parentheses. ***, ** and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

	<i>R&D Intensity</i>	<i>Ln(Patents)</i>	<i>Book Leverage</i>	<i>Operating Leverage</i>	<i>Ln(Segments)</i>
	(1)	(2)	(3)	(4)	(5)
Panel A: Base regressions					
<i>Constant</i>	0.128*** (0.000)	−2.951*** (0.000)	0.116 (0.235)	3.003 (0.134)	−1.006** (0.035)
<i>GAI</i>	0.006*** (0.000)	0.107** (0.045)	0.001 (0.868)	0.087 (0.539)	0.023 (0.337)
<i>GAI</i> $\times$ <i>Ln(Tenure)</i>	−0.002*** (0.007)	−0.038 (0.153)	0.002 (0.469)	−0.118* (0.092)	0.019 (0.127)
<i>Ln(Tenure)</i>	0.002** (0.032)	−0.020 (0.467)	0.006** (0.029)	0.011 (0.863)	−0.033** (0.023)
<i>Control Variables</i>	Yes	Yes	Yes	Yes	Yes
<i>Industry Fixed Effects</i>	Yes	Yes	Yes	Yes	Yes
<i>Year Fixed Effects</i>	Yes	Yes	Yes	Yes	Yes
Observations	13,225	10,046	13,225	9400	10,727
Adjusted <i>R</i> ²	0.514	0.577	0.374	0.034	0.338
Panel B: Industry $\times$ year fixed effects regressions					
	(1)	(2)	(3)	(4)	(5)
<i>Constant</i>	0.125*** (0.000)	−3.194*** (0.000)	0.249* (0.065)	1.988 (0.335)	−0.942 (0.149)

(Continues)

TABLE 5 (Continued)

	<i>R&D Intensity</i>	<i>Ln(Patents)</i>	<i>Book Leverage</i>	<i>Operating Leverage</i>	<i>Ln(Segments)</i>
	(1)	(2)	(3)	(4)	(5)
<i>GAI</i>	0.005*** (0.000)	0.106* (0.059)	0.001 (0.902)	0.093 (0.524)	0.016 (0.512)
<i>GAI</i> × <i>Ln(Tenure)</i>	−0.002** (0.017)	−0.038 (0.173)	0.003 (0.345)	−0.139* (0.053)	0.022* (0.094)
<i>Ln(Tenure)</i>	0.001* (0.080)	−0.024 (0.407)	0.005* (0.075)	0.004 (0.952)	−0.033** (0.029)
<i>Control Variables</i>	Yes	Yes	Yes	Yes	Yes
<i>Industry</i> × <i>Year Fixed Effects</i>	Yes	Yes	Yes	Yes	Yes
Observations	13,225	10,046	13,225	9400	10,727
Adjusted R^2	0.505	0.565	0.375	0.027	0.325

Abbreviations: CEO, chief executive officer; GAI, General Ability Index.

transportation and utilities firms (SIC codes 4000–4999) and firms that operate in four-digit SIC industries that, according to the extended KPSS data set, have never filed any patents. The control variables in this analysis include the baseline set and *CAPEX*, which is a firm's capital expenditure scaled by the total assets. The results in Column (2) show that the productivity of R&D expenses is generally consistent with R&D intensity: The coefficients on *GAI* are significantly positive, and the coefficients on *GAI* × *Ln(Tenure)* are negative, with marginal significance. In untabulated results, we also follow Islam and Zein (2020) in restricting the sample to high-tech firms as classified by Loughran and Ritter (2004)⁸ and run similar regressions. We obtain qualitatively similar results with the alternative sample.

Then, we examine a firm's financial policy and use *Book Leverage* as the dependent variable in Column (3) of Table 5. Higher-levered firms are associated with greater earnings volatility; thus, these firms have riskier financial policies than their lower-levered counterparts. The control variables in this specification are the baseline set excluding *Book Leverage*. The results in Column (3) indicate that CEOs' general managerial skills have no meaningful effect on firms' financial policy: Neither the coefficients on *GAI* nor those on *GAI* × *Ln(Tenure)* are statistically significant.

In Column (4) of Table 5, we consider a firm's policy of the utilization of fixed costs versus variable costs and use *Operating Leverage* as the dependent variable. A higher proportion of fixed costs increases the sensitivity of a firm's cash flow to market risk and raises the firm's systematic risk. Similar to Serfling (2014), *Operating Leverage* is measured as the elasticity of a

⁸Loughran and Ritter (2004) define high-tech firms as those in the industries of computer hardware (SIC codes 3571, 3572, 3575, 3577 and 3578), communications equipment (3661, 3663 and 3669), electronics (3671, 3672, 3674, 3675, 3677, 3678 and 3679), navigation equipment (3812), measuring and controlling devices (3823, 3825, 3826, 3827 and 3829), medical instruments (3841 and 3845), telephone equipment (4812 and 4813), communication services (4899) and software (7371, 7372, 7373, 7374, 7375, 7378 and 7379).

firm's operating income with respect to its sales. Conceptually, if a firm has no fixed costs, its *Operating Leverage* should equal one. A greater value of the variable indicates a higher proportion of fixed costs incurred in the firm, and thus a higher level of firm risk. The control variables for this specification are the baseline set plus *Labour Stock*, which is also a control variable in H. J. Chen et al. (2011), defined as the number of a firm's employees scaled by total assets. The results in Column (4) illustrate a weakening effect of the GAI with tenure on *Operating Leverage*: The coefficients on *GAI* are positive but statistically insignificant, whereas the coefficients on $GAI \times Ln(Tenure)$ are significantly negative, indicating that, relative to specialists, generalist CEOs use fewer fixed costs relative to variable costs the longer they manage their firms.

In Column (5) of Table 5, we consider a firm's diversification strategy and use $Ln(Segments)$ as the dependent variable. More diversified firms are less sensitive to market- or industry-wide shocks; thus, firms with more business segments are associated with lower risk. The results in Column (5) reveal that the effect of CEOs' managerial skills on firms' diversification strategy strengthens with CEO tenure, since the coefficients on $GAI \times Ln(Tenure)$ are positive and statistically significant in the specification that controls for industry $\times$ year fixed effects.

To measure a firm's diversification strategy more directly, we investigate the firm's M&A activities and use acquisitiveness, particularly the acquisitiveness of targets from unrelated industries, as a proxy for the degree of a firm's diversification. The M&A data are obtained from the SDC Mergers and Acquisitions database. We follow the literature (e.g., Gormley & Matsa, 2016; Netter et al., 2011) and screen the raw SDC sample by imposing the following requirements: (1) the deals must be completed, (2) 50% or more total shares are acquired in the transaction, (3) 90% or more of the total shares are owned by the acquirer after the transaction and (4) the ultimate parent of the acquirer and target are not the same (such that consolidations within holding companies or buybacks are excluded). We use *Total No. of M&As* and *No. of Diversifying M&As*, which measure the total number of M&As and the number of diversifying M&As that are announced in the fiscal year, respectively, as dependent variables. Deals whose acquirer and target do not share the same three-digit SIC code are identified as diversifying M&As.⁹ The OLS regression results of this analysis are reported in Table 6. The control variables are the baseline set plus *CAPEX* and *Free Cash Flow*, variables that are also controlled by Cain and McKeon (2016).

The results reported in Table 6 show that, consistent with the earlier finding that the effect of the GAI on the number of firm segments increases with tenure, the effect of the GAI on the number of acquisitions particularly the number of diversifying acquisitions also strengthens with tenure. In Columns (3)–(4) and (7)–(8), the coefficients on $GAI \times Ln(Tenure)$ are significantly positive. It is argued that, because the broader scope of knowledge and expertise equips generalist CEOs with abilities to select targets from unrelated industries and manage conglomerate firms, they are more likely to engage in diversifying M&As (G. Chen et al., 2020). If this dimension of general managerial skills dominates, we expect to observe a positive effect of CEOs' general managerial skills on firms' acquisitiveness of unrelated targets throughout the CEO tenure. However, the results shown in Columns (7) and (8) illustrate that, relative to specialists, generalists conduct fewer diversifying M&As in the earlier stage of their tenure and more diversifying M&As in the later stage of tenure. And, as Columns (5) and (6) show, the

⁹We use alternative definitions of diversifying M&As, such as deals whose acquirer and target do not share the same two-digit SIC code or the same four-digit SIC code, and obtain qualitatively similar results.

TABLE 6 CEOs' general managerial skills, tenure and M&As

This table presents regression results of the relation between CEOs' general managerial skills and the acquisitiveness of firms and the evolution of the relation with CEOs' tenure. The sample consists of Compustat ExecuComp firms from 1993 to 2007 whose CEOs' prior professional experience can be identified in the BoardEx database. The dependent variable in Columns (1)–(4) is the total number of M&As announced in a fiscal year (*Total No. of M&As*) and in Columns (5)–(8) is the number of diversifying M&As announced in a fiscal year (*No. of Diversifying M&As*). Deals in which the acquirer and target do not share the same three-digit SIC code are identified as diversifying M&As. The key independent variable is *GAI*, which is an index constructed by Custódio et al. (2013) as a measure of a CEO's general managerial ability. The mediating variable *Ln(Tenure)* is the natural logarithm of *Tenure* plus one. The control variables include *Ln(CEO Age)*, *Ln(CEO Delta)*, *Ln(CEO Vega)*, *Female Dummy*, *Ln(Assets)*, *Book Leverage*, *Tobin's Q*, *Asset Tangibility*, *R&D Intensity*, *ROA*, *Sales Growth*, *Ln(Firm Age)*, *Free Cash Flow* and *CAPEX*. The industry fixed effects are determined based on the firm's two-digit SIC code. The definitions of the variables are provided in Table A1 in Appendix A. All continuous variables are winsorized at their first and 99th percentiles. Standard errors are clustered by firm, and heteroskedasticity robust *p* values are provided in parentheses. ***, ** and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

	<i>Total No. of M&As</i>				<i>No. of Diversifying M&As</i>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Constant</i>	−0.997 (0.176)	−0.586 (0.421)	−1.029 (0.161)	−0.633 (0.384)	−0.364 (0.525)	−0.239 (0.674)	−0.384 (0.503)	−0.270 (0.635)
<i>GAI</i>	−0.005 (0.826)	−0.011 (0.675)	−0.106** (0.016)	−0.105** (0.023)	0.002 (0.911)	0.002 (0.924)	−0.061* (0.071)	−0.059* (0.096)
<i>GAI × Ln(Tenure)</i>			0.059** (0.014)	0.055** (0.026)			0.037** (0.028)	0.036** (0.046)
<i>Ln(Tenure)</i>	−0.019 (0.458)	−0.018 (0.476)	−0.017 (0.498)	−0.017 (0.509)	−0.027 (0.119)	−0.024 (0.182)	−0.026 (0.135)	−0.023 (0.198)
<i>Control Variables</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Industry Fixed Effects</i>	Yes	No	Yes	No	Yes	No	Yes	No
<i>Year Fixed Effects</i>	Yes	No	Yes	No	Yes	No	Yes	No
<i>Industry × Year Fixed Effects</i>	No	Yes	No	Yes	No	Yes	No	Yes
Observations	11,583	11,583	11,583	11,583	11,583	11,583	11,583	11,583
Adjusted <i>R</i> ²	0.107	0.112	0.108	0.112	0.076	0.073	0.077	0.074

Abbreviations: CEO, chief executive officer; GAI, General Ability Index.

overall effect of the *GAI* on the *No. of Diversifying M&As* is positive but statistically insignificant. These results suggest that generalists' engagement in the M&As of unrelated targets is not essentially driven by their broad scope of expertise. Rather, the results are more consistent with the notion that, because they have more outside job opportunities relative to specialists, generalists are less risk averse and take on more risk (conduct fewer diversifying M&As) in the earlier stage of their tenure; and, due to the general to firm-specific managerial skills transformation, generalist CEOs become more risk averse and tend to undertake less risky policies (conduct more diversifying M&As) the longer they are tenured with the firm.

Taken together, in this subsection, we examine the effect of the GAI on the riskiness of corporate policies and the evolution of the effect with CEO tenure. Except for *Book Leverage*, on which the GAI has no meaningful effect, we find that the effect of the GAI on the riskiness of policies weakens with tenure. Specifically, generalists tend to spend less on R&D, use fewer fixed costs relative to variable costs, and implement a higher level of diversification strategy the longer they manage the firm. There is also evidence suggesting that generalists tend to undertake riskier policies than specialists in the early stage of their tenure. For example, generalists are more aggressive in R&D policy and patent applications. These results are consistent with our main findings and demonstrate that CEOs' general managerial skills affect firm risk through their impact on the riskiness of corporate policies.

3.3 | Mechanism: General to firm-specific managerial skills transformation

In the earlier subsections, we find CEOs' general managerial skills to have a significantly positive effect on firm risk in the early stage of the CEO tenure and a pattern in which the effect weakens with tenure. We conjecture that the findings are attributable to a transformation from general to firm-specific managerial skills of CEOs as follows: generalist CEOs' managerial skills become more firm specific over time, rendering them more risk averse the longer they manage the firms. If this conjecture is true, we expect that the effect of CEO tenure on the general managerial skills–firm risk relation should differ across two groups of CEOs, that is, insiders and outsiders. Outsiders are those who are appointed as CEOs shortly after joining the firm. After they are appointed, their managerial skills are expected to gradually become vested in the firm over time. In contrast, insiders have been with the firm for a relatively long time before being appointed as CEOs, and their skills have already, to some extent, become firm specific by the time they are appointed. Therefore, if generalist CEOs' risk-taking behaviour is closely associated with the extent to which their skills are vested in the firm, we expect that the pattern of tenure weakening their risk-taking behaviour should be more pronounced among outsiders than among insiders. To ensure that the insiders' skills have been vested in the firm to a meaningful degree before these insiders are appointed as CEOs, we follow Brockman et al. (2016) and define insiders as CEOs whose preappointment tenure is above the sample median (i.e., 4 years), and the other CEOs as outsiders.¹⁰

Table 7 reports the OLS regression results of this analysis in which the total sample is split into two subsamples: one sample consisting of outsiders and the other consisting of insiders. Then, we run the same regressions as specifications (2) and (5) of Table 3 for each subsample. The dependent variable is *Total Risk* and the control variables are from the baseline set. Columns (1) and (2) include separate industry and year fixed effects, and Columns (3) and (4) include industry $\times$ year fixed effects. The results reveal that our main findings are indeed more pronounced among outsiders than among insiders. In the regression results for the outsiders (Columns 1 and 3), the coefficients on *GAI* are significantly positive and the coefficients on $GAI \times Ln(Tenure)$ are significantly negative. However, in the regression results for the insiders

¹⁰In untabulated results, we use alternative definitions of outsiders (e.g., CEOs with a preappointment tenure of less than 2 or 3 years) or those of insiders (e.g., CEOs with a preappointment tenure in the top quartile of the sample, that is, more than 16 years) and find that the alternative definitions do not qualitatively change the results.

TABLE 7 Results of outsiders versus insiders

This table compares outsiders and insiders in terms of the relation between CEOs' general managerial skills and firm risk and the evolution of the relation with CEOs' tenure. Outsiders (insiders) are defined as CEOs whose preappointment tenure is equal to or less (greater) than the sample median (i.e., 4 years). The dependent variable is *Total Risk*. The key independent variable is *GAI*, which is an index constructed by Custódio et al. (2013) as a measure of a CEO's general managerial ability. The mediating variable $\ln(\text{Tenure})$ is the natural logarithm of *Tenure* plus one. The control variables include $\ln(\text{CEO Age})$, $\ln(\text{CEO Delta})$, $\ln(\text{CEO Vega})$, *Female Dummy*, $\ln(\text{Assets})$, *Book Leverage*, *Tobin's Q*, *Asset Tangibility*, *R&D Intensity*, *ROA*, *Sales Growth* and $\ln(\text{Firm Age})$. The industry fixed effects are determined based on the firm's two-digit SIC code. The definitions of the other variables are provided in Table A1 in Appendix A. All continuous variables are winsorized at their first and 99th percentiles. Standard errors are clustered by firm, and heteroskedasticity robust *p* values are provided in parentheses. ***, ** and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

	Outsiders (1)	Insiders (2)	Outsiders (3)	Insiders (4)
<i>Constant</i>	1.123*** (0.000)	0.850*** (0.000)	1.162*** (0.000)	0.912*** (0.000)
<i>GAI</i>	0.018** (0.016)	−0.002 (0.848)	0.023*** (0.003)	0.003 (0.735)
$GAI \times \ln(\text{Tenure})$	−0.008** (0.024)	−0.002 (0.686)	−0.010** (0.011)	−0.005 (0.248)
$\ln(\text{Tenure})$	0.009* (0.062)	0.001 (0.703)	0.012** (0.026)	0.003 (0.555)
<i>Control Variables</i>	Yes	Yes	Yes	Yes
<i>Industry Fixed Effects</i>	Yes	Yes	No	No
<i>Year Fixed Effects</i>	Yes	Yes	No	No
<i>Industry × Year Fixed Effects</i>	No	No	Yes	Yes
Observations	3564	3457	3564	3457
Adjusted R^2	0.625	0.600	0.649	0.628
Chow test				
$GAI \times \text{Outsider Dummy}$	0.020** (0.015)		0.020** (0.016)	
$GAI \times \ln(\text{Tenure}) \times \text{Outsider Dummy}$	−0.007* (0.088)		−0.004 (0.273)	

Abbreviations: CEO, chief executive officer; GAI, General Ability Index.

(Columns 2 and 4), neither the coefficients on *GAI* nor those on $GAI \times \ln(\text{Tenure})$ are statistically significant.

To further examine whether the estimated coefficients on *GAI* and (or) $GAI \times \ln(\text{Tenure})$ statistically differ between the two subsamples, we conduct Chow tests for the pooled sample of outsiders and insiders. Specifically, we introduce an *Outsider Dummy*, which is a binary

variable that equals one if the CEO is an outsider, and zero otherwise, and interact this variable with all independent variables in the model of Table 7 (including GAI , $GAI \times Ln(Tenure)$, $Ln(Tenure)$, all the control variables and the fixed effects). Then, we regress *Total Risk* on all the independent variables, *Outsider Dummy* and the interaction terms between *Outsider Dummy* and the independent variables. Therefore, sufficient flexibility is allowed for the model to capture outsider–insider differences.¹¹ In Table 7, we report only the estimated coefficients on our variables of interest, $GAI \times Outsider Dummy$ and $GAI \times Ln(Tenure) \times Outsider Dummy$, which capture the outsider–insider differences in the effects on *Total Risk* of GAI and $GAI \times Ln(Tenure)$, respectively. Generally, consistent with the subsample regression results, the results of the Chow tests show that our main findings are more pronounced for outsiders than for insiders, since the coefficients on $GAI \times Outsider Dummy$ are significantly positive in both models and the coefficient on $GAI \times Ln(Tenure) \times Outsider Dummy$ is significantly negative in the first model, which controls for industry and year fixed effects separately. On the other hand, it seems that the outsider–insider differences in terms of the tenure weakening the effect of GAI on *Total Risk* are statistically weak (in particular, the estimated coefficient on $GAI \times Ln(Tenure) \times Outsider Dummy$ is negative but statistically insignificant in the second Chow test model that includes the industry $\times$ year fixed effects). Although a potentially serious multicollinearity problem could have made the Chow test models largely underestimated the significance of the estimated coefficients,¹² including those on $GAI \times Ln(Tenure) \times Outsider Dummy$, the Chow test results suggest us be more cautious when interpreting our main findings.

Taken together, Table 7 shows that our main findings, the positive effect of CEOs' general managerial skills on firm risk in the early stage of CEO tenure and the pattern of tenure weakening the effect, are more pronounced among outsiders, whose human capital has not yet been vested in the firm when they were appointed as CEOs, than among insiders, whose human capital was largely firm specific when they were appointed. These results are consistent with our conjectured explanation of the main findings: A general to firm-specific transformation of CEOs' managerial skills occurs as these CEOs' tenure with the firm increases.

4 | ENDOGENEITY ISSUE

Given our main results' pattern of varying with tenure, some typical endogeneity problems (e.g., reverse causality) can be ruled out. For example, because firm risk is decreased in the later period of generalist CEOs' tenure, firms are unlikely to recruit generalists to shift to a riskier strategy. Additionally, firms are unlikely to recruit generalist CEOs to change to a less risky strategy. First, both economic intuition and empirical evidence (e.g., Custódio et al., 2019; Mishra, 2014) suggest that generalists tend to be more risk loving than average CEOs. In

¹¹Assuming that the effects on firm risk of the control variables and the fixed effects are the same between the two subsamples, using the pooled sample, we also run the models in which *Total Risk* is regressed on GAI , $GAI \times Ln(Tenure)$, $Ln(Tenure)$, *Outsider Dummy* and its respective interactions with GAI , $GAI \times Ln(Tenure)$, $Ln(Tenure)$, the baseline control variables and the fixed effects. We obtain results qualitatively similar to those of the Chow tests.

¹²In the first (second) model with firm and year fixed effects (industry $\times$ year fixed effects), the variance inflation factor (VIF) of $GAI \times Outsider Dummy$ is 18.17 (21.26), the VIF of $GAI \times Ln(Tenure) \times Outsider Dummy$ is 16.61 (19.49) and the average VIF of the independent variables is 59.25 (17.30), all of which are greater than the threshold of 10 for exhibiting high multicollinearity (e.g., Belsley et al., 1980).

addition, it takes too long (more than 3 years) for generalists to reduce firm risk. However, as discussed later, an endogeneity problem can arise from nonrandom CEO–firm matching, that is, some omitted factors could affect both CEOs' human capital generality and firm risk. In the next three subsections, we use the methods of PSM, firm or CEO fixed effects and IV to address the potential endogeneity concern arising from omitted variables.

4.1 | PSM method

We first use the PSM method to address the concern about endogeneity arising from the potential omission of observable variables. Specifically, we compare the risk of firms run by generalists (treated group) with the risk of firms run by specialists with very similar characteristics (control group) and particularly examine how the difference in firm risk between the two groups changes with CEO tenure. An advantage of the PSM method is that it can eliminate a number of observable differences between the treated and control groups.

While constructing the matching sample, we match firm–years with newly recruited generalists and firm–years with newly recruited specialists on several firm characteristics as of 1 year before the recruitment of the CEOs. Specifically, we first measure the propensity scores by running a logit model in which the dependent variable is *Generalist Dummy*, which is a binary variable that equals one if the firm recruited a generalist, and zero otherwise, and the independent variables include *Ln(Assets)*, *Book Leverage*, *Tobin's Q*, *Asset Tangibility*, *R&D Intensity*, *ROA*, *Sales Growth* and *Ln(Firm Age)* as of 1 year before the CEO recruitment.¹³ After obtaining the scores, we construct the matching sample using the nearest-neighbour score matching technique. Following Mishra (2014), we construct two sets of matched samples with and without restrictions on the repeated matching of the control firm–years. To save space, we report only the results of the matched sample of nonrepeated control firm–years.¹⁴ The statistics reported in Panel A of Table 8 show that the differences in firm characteristics between the treated and control groups are largely eliminated. Except for *Ln(Assets)*, which shows a treated–control difference at the 10% level of significance, the differences in the other variables between the two groups are negligible and statistically insignificant.

After obtaining the matched sample, we compare *Total Risk* between the treated and control groups in the years after the CEO recruitment. To determine how the difference in firm risk changes with CEO tenure, we divide the matched sample based on the median of *Tenure* of the original total sample (i.e., 5 years) and, for each subsample, conduct a comparison. Panel B of Table 8 reports the statistics, which show a pattern consistent with our main findings: The difference in *Total Risk* between the treated and control groups is significantly positive in the subsample with CEO tenure within the median and becomes insignificant in the subsample with above-median tenure. To alleviate the concern that some firm variables could become very different between the two groups after matching, we conduct a multivariate analysis, which also allows us to control for CEO characteristics and fixed effects. For each subsample, we regress *Total Risk* on *Treated Group Dummy* (a binary variable indicating that the firm–year is

¹³We also use alternative sets of independent variables—for example, we closely follow Custódio et al. (2013) and use *Ln(Assets)*, *Book Leverage*, *R&D Intensity*, *ROA* and *Ln(Segments)* as of 1 year before the recruitment—and obtain qualitatively similar results.

¹⁴The results obtained from the matching sample without restrictions on repeated control firms are qualitatively similar.

TABLE 8 PSM results

This table presents the results of using the PSM method in which the firm risk of the treated group (firms run by generalists) and the control group (firms run by specialists) is compared along CEOs' tenure. Firm-years with newly hired generalist CEOs are matched with firm-years with newly hired specialist CEOs using the nearest-neighbour score matching technique. The matching scores are generated by a logit model in which the dependent variable is a binary variable indicating that the firm recruited a generalist CEO, and the independent variables include *Ln(Assets)*, *Book Leverage*, *Tobin's Q*, *Asset Tangibility*, *R&D Intensity*, *ROA*, *Sales Growth* and *Ln(Firm Age)* as of 1 year before the CEO recruitment. The results reported in this table are based on the matched sample with nonrepeated control firm-years. Panel A reports the statistics of the firm characteristics of the matched sample. The mean values of each variable are presented and compared between the treated and control groups. Panel B reports the difference in *Total Risk* between the treated and control groups in the years after the CEO recruitment; in particular, *Total Risk* is compared in two subsamples: *Tenure* ≤ 5 and *Tenure* > 5. Panel C reports the regression results of the multivariate analysis of the treated-control difference in *Total Risk* for the two tenure groups. The dependent variable is *Total Risk*. The independent variable of interest is *Treated Group Dummy*, which is a binary variable that equals one if the firm-year is in the treated group, and zero otherwise. The control variables include *Ln(Tenure)*, *Ln(CEO Age)*, *Ln(CEO Delta)*, *Ln(CEO Vega)*, *Female Dummy*, *Ln(Assets)*, *Book Leverage*, *Tobin's Q*, *Asset Tangibility*, *R&D Intensity*, *ROA*, *Sales Growth* and *Ln(Firm Age)*. The industry fixed effects are determined based on the firm's two-digit SIC code. The definitions of all variables are provided in Table A1 in Appendix A. All continuous variables are winsorized at their first and 99th percentiles. Standard errors are clustered by firm, and heteroskedasticity robust *p* values are provided in the parentheses. ***, ** and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

Panel A: Statistics on firm characteristics for the matched sample				
	Treated	Control	Treated-control difference	T statistics
<i>Ln(Assets)</i>	7.475	7.640	−0.165	−1.90*
<i>Book Leverage</i>	0.222	0.221	0.001	0.14
<i>Tobin's Q</i>	1.987	1.978	0.009	0.11
<i>Asset Tangibility</i>	0.279	0.286	−0.007	−0.59
<i>R&D Intensity</i>	0.031	0.030	0.001	0.30
<i>ROA</i>	0.079	0.082	−0.003	−0.50
<i>Sales Growth</i>	0.108	0.120	−0.012	−0.67
<i>Ln(Firm Age)</i>	2.946	2.939	0.007	0.16
Panel B: Statistics on Total Risk for the matched sample				
Subsample	Treated	Control	Difference	T statistics
<i>Tenure</i> ≤ 5	0.40	0.38	0.02	4.30***
<i>Tenure</i> > 5	0.32	0.34	−0.02	−1.51
Panel C: Multivariate analysis on the treated-control difference in Total Risk				
	Dependent variable: Total Risk			
	<i>Tenure</i> ≤ 5		<i>Tenure</i> > 5	
<i>Constant</i>	0.718*** (0.000)	0.840*** (0.000)	0.334 (0.259)	0.714** (0.043)
<i>Treated Group Dummy</i>	0.016*** (0.005)	0.015** (0.015)	−0.010 (0.455)	−0.006 (0.680)

(Continues)

TABLE 8 (Continued)

Panel C: Multivariate analysis on the treated–control difference in <i>Total Risk</i>				
	Dependent variable: <i>Total Risk</i>			
	<i>Tenure</i> ≤ 5		<i>Tenure</i> > 5	
<i>Control Variables</i>	Yes	Yes	Yes	Yes
<i>Industry Fixed Effects</i>	Yes	No	Yes	No
<i>Year Fixed Effects</i>	Yes	No	Yes	No
<i>Industry × Year Fixed Effects</i>	No	Yes	No	Yes
Observations	3312	3312	557	557
Adjusted R^2	0.600	0.627	0.553	0.596

Abbreviations: CEO, chief executive officer; PSM, propensity score matching; SIC, Standard Industrial Classification.

in the treated group), control variables (including the baseline set and $\ln(\textit{Tenure})$) and fixed effects. The subsample regression results reported in Panel C of Table 8 also present a consistent pattern: The coefficients on *Treated Group Dummy* are significantly positive in the subsample with early CEO tenure and insignificant in the subsample with later tenure.

4.2 | Firm or CEO fixed effects

Unobservable omitted variables can be time varying or time invariant. For example, suppose an unobservable and time-decaying factor z_1 has a positive and constant effect on both the key independent variable (*GAI*) and the dependent variable (*Total Risk*). The omission of z_1 will lead to the pattern of our main finding: a weakening effect of *GAI* on *Total Risk* with time. Alternatively, if an unobservable variable z_2 is time invariant but its relation with either *GAI*, *Total Risk* or both weakens with time, the omission of z_2 could also lead to the pattern of our main finding. In this subsection, we use the firm or CEO fixed effects method to address the problem of omitted unobservable time-invariant variables. Then, in Section 4.3, we use the IV method to address the problem of omitted unobservable time-varying variables.

Table 9 reports the regression results obtained after controlling for firm or CEO fixed effects. In Column (1) ([2]), *Total Risk* is regressed on *GAI*, $\textit{GAI} \times \ln(\textit{Tenure})$, $\ln(\textit{Tenure})$, baseline controls, year fixed effects and firm (CEO) fixed effects. The results in Table 9 suggest that our main findings are robust to the inclusion of firm or CEO fixed effects: in both Columns (1) and (2), the coefficients on *GAI* are significantly positive and the coefficients on $\textit{GAI} \times \ln(\textit{Tenure})$ are significantly negative, indicating that our main findings are not attributable to omitted unobservable time-invariant variables.

4.3 | IV method

Subsequently, we use the IV method to address the concern about potential endogeneity arising from omitted unobservable time-varying variables. Following the related literature (e.g., Fisman & Svensson, 2007; Lin et al., 2012), we instrument CEOs' general managerial skills

TABLE 9 Results of firm or CEO fixed effects

This table presents the regression results of the relation between CEOs' general managerial skills and firm risk and the evolution of the relation with CEOs' tenure after controlling for firm or CEO fixed effects. The dependent variable is *Total Risk*, which is defined as the annualized daily stock return in a fiscal year. The key independent variable is *GAI*, which is an index constructed by Custódio et al. (2013) as a measure of a CEO's general managerial ability. The mediating variable $\ln(\textit{Tenure})$ is the natural logarithm of *Tenure* plus one. The control variables include $\ln(\textit{CEO Age})$, $\ln(\textit{CEO Delta})$, $\ln(\textit{CEO Vega})$, *Female Dummy*, $\ln(\textit{Assets})$, *Book Leverage*, *Tobin's Q*, *Asset Tangibility*, *R&D Intensity*, *ROA*, *Sales Growth* and $\ln(\textit{Firm Age})$. The definitions of the other variables are provided in Table A1 in Appendix A. All continuous variables are winsorized at their first and 99th percentiles. Standard errors are clustered by firm, and heteroskedasticity robust *p* values are provided in parentheses. ***, ** and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

	(1)	(2)
<i>Constant</i>	0.809*** (0.000)	2.123*** (0.000)
<i>GAI</i>	0.006* (0.053)	0.021*** (0.000)
$\textit{GAI} \times \ln(\textit{Tenure})$	−0.003* (0.095)	−0.005** (0.019)
$\ln(\textit{Tenure})$	0.004** (0.018)	0.006* (0.061)
<i>Control Variables</i>	Yes	Yes
<i>Firm Fixed Effects</i>	Yes	No
<i>CEO Fixed Effects</i>	No	Yes
<i>Year Fixed Effects</i>	Yes	Yes
Observations	12,797	12,797
Adjusted R^2	0.736	0.763

Abbreviations: CEO, chief executive officer; GAI, General Ability Index.

using the average GAI of comparison groups with similar characteristics. The justification for such instruments is that the group average of the GAI is correlated with the endogenous regressor *GAI* but is not otherwise related to firm risk. However, this method could be problematic. As criticized by Gormley and Matsa (2014), using the same industry group as the comparison group, if an endogeneity problem arises due to the omission of unobserved industry heterogeneity that affects both the endogenous independent variable (and thus the IV) and the dependent variable, the exclusion restriction could be violated and lead to inconsistent IV estimation. The inclusion of industry fixed effects might not solve the problem, since the demeaned IV could be perfectly correlated with the endogenous regressor in the first stage; thus, the model is not well identified.¹⁵

¹⁵See the detailed argument on pages 635–636 in Gormley and Matsa (2014).

To alleviate the concern raised by Gormley and Matsa (2014), we include industry fixed effects or industry $\times$ year fixed effects in the second stage to control for the potential unobserved industry heterogeneity that affects both the IV and the dependent variable to avoid violating the exclusion restrictions. Meanwhile, to maintain the relevance condition of the IV, we attempt to introduce to the comparison group other variations beyond the narrow industry classification that are related to the endogenous regressor such that the demeaned IV by the narrow industry groups could still contain variations correlated with the endogenous regressor in the first stage. In this regard, we introduce to the comparison group the following two dimensions: firm size and a broader industry classification (i.e., a broad Fama–French industry classification)¹⁶; thus, the demeaned IV in the first stage contains firm size variation and variation from a broader industry and is expected to be positively correlated with the endogenous regressor, *GAI*. Specifically, each year, we divide the initial sample into 10 size groups based on the decile of the firms' total assets and construct variable *IV* as the mean *GAI* of the CEOs from firms in the same size group and the same Fama–French 12 industry group.¹⁷ As before, we use the firms' two-digit SIC code as the industry classification for industry fixed effects.¹⁸

Notably, our main regression model contains the interaction term $GAI \times Ln(Tenure)$. If *GAI* is endogenous, the interaction between *GAI* and $Ln(Tenure)$ is also endogenous. We follow the literature (e.g., Rajan & Zingales, 1998) and normal practice to construct $IV \times Ln(Tenure)$ and then run the following two regressions in the first stage: We regress *GAI* and $GAI \times Ln(Tenure)$, respectively, on *IV*, $IV \times Ln(Tenure)$, and all independent variables in the second stage. In the second stage, we use the predicted *GAI* and $GAI \times Ln(Tenure)$ from the first-stage regressions to instrument the endogenous *GAI* and $GAI \times Ln(Tenure)$ and estimate their effects on *Total Risk*.

Table 10 reports the results of the IV estimation. The control variables are from the baseline set. Columns (1)–(3) include industry and year fixed effects separately, and Columns (4)–(6) include industry $\times$ year fixed effects. The first-stage regression results (Columns 1, 2, 4 and 5) show that *GAI* and $GAI \times Ln(Tenure)$ are significantly correlated with the IVs. In particular, as predicted, *IV* is significantly and positively correlated with *GAI*. The Cragg–Donald Wald *F* statistics are 295.19 and 286.23, which are much higher than the Stock–Yogo 10% critical value (7.03) to reject weak IVs. The second-stage regression results (Columns 3 and 6) are consistent with our main findings: The coefficients on the fitted value of the *GAI* are significantly positive, and the coefficients on the fitted $GAI \times Ln(Tenure)$ are significantly negative.

Taken together, the results of the PSM, firm or CEO fixed effects and IV methods indicate that our main findings are robust to endogeneity concerns. Considering the potential nonrandom CEO–firm matching arising from omitted variables, we still observe a positive effect of CEOs' general managerial skills on firm risk in the early stage of CEO tenure and a pattern of the effect weakening with tenure.

¹⁶According to Custódio et al. (2013), Mishra (2014) and the untabulated results of the logit model of the PSM approach in this paper, firm size is significantly and positively related to a firm's recruitment of generalist CEOs. Moreover, Mishra (2014) contends that firms' Fama–French industry membership is related to their recruitment of generalists ("some industries (e.g., retail) are more likely to feature a generalist CEO compared with other industries (e.g., high-tech)", page 399).

¹⁷We also attempt to divide the initial total sample into 4 size groups based on the quartile points of the firms' total assets or (and) use the Fama–French 17 broad industry classifications. The results are qualitatively the same.

¹⁸Our sample contains 44 unique two-digit SIC groups. Thus, the Fama–French 12 and 17 industry classifications are roughly broader than the two-digit-SIC industry classification.

TABLE 10 Instrumental variable

This table presents the estimates of the IV method. Each year, the total sample is divided into 10 size groups based on the deciles of firms' total assets, and *IV* is constructed as the mean *GAI* of the CEOs from firms in the same size group and the same Fama–French 12 industry group. In the first stage, *GAI* and $GAI \times Ln(Tenure)$ are regressed on *IV*, $IV \times Ln(Tenure)$, $Ln(Tenure)$, the control variables and fixed effects of the second stage. In the second stage, *Total Risk* is regressed on *GAI* (fitted value) and $GAI \times Ln(Tenure)$ (fitted value) predicted from the first stage, $Ln(Tenure)$, the control variables and fixed effects. The control variables include $Ln(CEO\ Age)$, $Ln(CEO\ Delta)$, $Ln(CEO\ Vega)$, *Female Dummy*, $Ln(Assets)$, *Book Leverage*, *Tobin's Q*, *Asset Tangibility*, *R&D Intensity*, *ROA*, *Sales Growth* and $Ln(Firm\ Age)$. The Fixed effects included in Columns (1)–(3) are the industry and year fixed effects and in Columns (4)–(6) are industry $\times$ year fixed effects. The industry fixed effects are determined based on the firm's two-digit SIC code. The definitions of all variables are provided in Table A1 in Appendix A. All continuous variables are winsorized at their first and 99th percentiles. Standard errors are clustered by firm, and heteroskedasticity robust *p* values are provided in parentheses. ***, ** and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

	First stage		Second stage	First stage		Second stage
	<i>GAI</i>	$GAI \times Ln(Tenure)$		<i>GAI</i>	$GAI \times Ln(Tenure)$	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Constant</i>	−5.592*** (0.000)	−8.254*** (0.000)	1.036*** (0.000)	−5.513*** (0.000)	−8.479*** (0.000)	1.074*** (0.000)
<i>IV</i>	0.482*** (0.000)	−0.722*** (0.000)		0.519*** (0.000)	−0.666*** (0.000)	
$IV \times Ln(Tenure)$	0.139*** (0.000)	1.158*** (0.000)		0.134*** (0.000)	1.159*** (0.000)	
<i>GAI</i> (fitted value)			0.045*** (0.000)			0.048*** (0.000)
$GAI \times Ln(Tenure)$ (fitted value)			−0.010** (0.046)			−0.012** (0.012)
$Ln(Tenure)$	−0.094*** (0.000)	−0.211*** (0.000)	0.008*** (0.002)	−0.096*** (0.000)	−0.215*** (0.000)	0.008*** (0.002)
<i>Control Variables</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Industry Fixed Effects</i>	Yes	Yes	Yes	No	No	No
<i>Year Fixed Effects</i>	Yes	Yes	Yes	No	No	No
<i>Industry $\times$ Year Fixed Effects</i>	No	No	No	Yes	Yes	Yes
Observations	11,219	11,219	11,219	11,219	11,219	11,219
Adjusted R^2	0.247	0.269	0.606	0.234	0.255	0.641
Cragg–Donald Wald <i>F</i> statistics for IV	295.186			286.227		

Abbreviations: CEO, chief executive officer; GAI, General Ability Index; IV, instrumental variable.

5 | RULING OUT ALTERNATIVE EXPLANATIONS

In this section, we explore alternative explanations other than the general to firm-specific managerial skills transformation hypothesis for our main findings. First, we investigate whether our main results are driven by changes in managerial preferences with tenure. Bertrand and Mullainathan (2003) find that when CEOs are not closely monitored, they are likely to lead a “quiet life”. In particular, the CEOs may forgo risky investments in R&D and reduce firm risk. Gormley and Matsa (2016) find that after CEOs are sheltered from the market for corporate control, they tend to “play it safe”, that is, to take on less risk that is desired by diversified shareholders or even undertake value-destroying actions that reduce the firm’s risk. With accumulated experiences, generalists are likely to be more powerful than specialists and, thus, associated with weaker governance (Custódio et al., 2013). If generalists tend to pursue a “quiet life” or “play it safe” in the late stage of their tenure, we expect to observe a weakening effect of CEOs’ general managerial skills on firm risk with the tenure.

We conduct two tests to investigate the above two hypotheses. First, both hypotheses illustrate aspects of managerial preferences that lead managers to act against the best interest of shareholders. Such managerial preferences manifest when the corporate governance is weak. Therefore, if either of the hypotheses dominates, our main finding of the tenure weakening the effect of the GAI on firm risk is supposed to be more pronounced among firms with weaker corporate governance. We follow the literature (e.g., Custódio et al., 2013) and use the Gompers et al. (2003) governance index (henceforth G index) as the measure of external corporate governance.¹⁹ The G index, which measures the number of the 24 governance provisions adopted by a firm, is obtained from the Institutional Shareholder Services (ISS) database.²⁰ A higher G index indicates a lower likelihood that a firm is taken over and the managers are replaced, and, hence, weaker external corporate governance. We divide our sample by the median of the G index (i.e., 9) and conduct a cross-sectional analysis with the two subsamples. The regression results reported in Panel A of Table 11 show that, instead of being more pronounced among the weak governance firm-years (the high G index group) as predicted by the hypotheses of the pursuit of “quiet life” or “playing it safe”, the weakening effect of the GAI on firm risk with the CEO tenure is actually more pronounced among strong governance firm-years (the low G index group). Second, both pursuing a “quiet life” and “playing it safe” are managerial preferences that are value destroying. If generalists tend to manifest these preferences in the later stage of their tenure, they are expected to be more value destroying in the later stage relative to the earlier stage of their tenure. We use risk-adjusted stock returns and M&A announcement returns as measures for firm value. *Risk-Adjusted Stock Return* is calculated as the annual stock return adjusted by the Fama–French three-factor model. The results reported in Columns (1) and (2) of Table 11 Panel B show that, in contrast to the prediction of the pursuit of “quiet life” or “playing it safe” hypotheses, the effect of the GAI on risk-adjusted stock returns strengthens with CEO tenure. Following Gormley and Matsa (2016), we calculate the M&A announcement return (denoted as *CAR5*) as the percentage cumulative abnormal return of the acquirer over $[-2, 2]$ days around the deal’s announcement date, with

¹⁹We also use Bebchuk et al. (2009) entrenchment index (E index) as an alternative measure for external corporate governance and obtain qualitatively similar results.

²⁰During our sample period, the ISS database published seven volumes in years 1993, 1995, 1998, 2000, 2002, 2004 and 2006. Following Gompers et al. (2003), we assume that during the years between two consecutive publications, firms have the same governance provisions as in the previous publication year.

TABLE 11 Ruling out alternative explanations

This table attempts to rule out explanations other than the “general to firm-specific” managerial skills transformation hypothesis. Panel A compares high G Index and low G Index firm-years in terms of the relation between CEOs’ general managerial skills and firm risk and the evolution of the relation with CEOs’ tenure. High (low) G Index indicates that the G index of the firm-year is above (below) the sample median (i.e., 9). Panel B reports regression results of the relation between CEOs’ general managerial skills and firm performance and the evolution of the relation with CEOs’ tenure. The dependent variables are *Risk-Adjusted Stock Return* (Columns 1 and 2) and *ROA* (Columns 3 and 4). Panel C examines whether the observed effects of the GAI on firm risk reflect the risk-taking pattern of outsiders. The key independent variable is *GAI_r*, which is constructed as the residuals from the regressions of *GAI* on an outsider dummy, *Ln(Tenure)*, baseline control variables (including *Ln(CEO Age)*, *Ln(CEO Delta)*, *Ln(CEO Vega)*, *Female Dummy*, *Ln(Assets)*, *Book Leverage*, *Tobin’s Q*, *Asset Tangibility*, *R&D Intensity*, *ROA*, *Sales Growth* and *Ln(Firm Age)*), and fixed effects. Outsiders are defined as CEOs whose preappointment tenure is below the sample median (i.e., 4 years) in Columns (1) and (2), and CEOs whose preappointment tenure is less than or equal to 1 year in Columns (3) and (4). Panel D examines the effects of CEO age. Columns (1) and (2) include *CEO Age × Year Fixed Effects*, and Columns (3) and (4) include *CEO Age Quintile × Year Fixed Effects*. The dependent variable in Panels A, C and D is *Total Risk*. The control variables include *Ln(CEO Age)*, *Ln(CEO Delta)*, *Ln(CEO Vega)*, *Female Dummy*, *Ln(Assets)*, *Book Leverage*, *Tobin’s Q*, *Asset Tangibility*, *R&D Intensity*, *ROA*, *Sales Growth* and *Ln(Firm Age)*. The industry fixed effects are determined based on the firm’s two-digit SIC code. The definitions of the variables are provided in Table A1 in Appendix A. All continuous variables are winsorized at their first and 99th percentiles. Standard errors are clustered by firm, and heteroskedasticity robust *p* values are provided in parentheses. ***, ** and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

Panel A: Cross-sectional analysis: By corporate governance				
	Dependent variable: Total Risk			
	High G Index	Low G Index	High G Index	Low G Index
	(1)	(2)	(3)	(4)
<i>Constant</i>	0.987*** (0.000)	0.748*** (0.000)	1.032*** (0.000)	0.970*** (0.000)
<i>GAI</i>	0.010* (0.060)	0.019*** (0.003)	0.011** (0.045)	0.019*** (0.006)
<i>GAI × Ln(Tenure)</i>	−0.003 (0.355)	−0.009*** (0.003)	−0.004 (0.177)	−0.009*** (0.005)
<i>Ln(Tenure)</i>	0.006 (0.132)	0.003 (0.413)	0.008* (0.091)	0.004 (0.297)
<i>Control Variables</i>	Yes	Yes	Yes	Yes
<i>Industry Fixed Effects</i>	Yes	Yes	No	No
<i>Year Fixed Effects</i>	Yes	Yes	No	No
<i>Industry × Year Fixed Effects</i>	No	No	Yes	Yes
<i>Observations</i>	4843	5026	4843	5026
<i>Adjusted R²</i>	0.562	0.605	0.581	0.630

Panel B: GAI, tenure and firm performance

	Dependent variable			
	Risk-Adjusted Stock Return		ROA	
	(1)	(2)	(3)	(4)
<i>Constant</i>	−0.283*** (0.000)	−0.317*** (0.000)	−0.064* (0.087)	−0.089*** (0.008)
<i>GAI</i>	−0.006*** (0.000)	−0.006*** (0.000)	−0.008*** (0.000)	−0.008*** (0.000)
<i>GAI × Ln(Tenure)</i>	0.003*** (0.001)	0.003*** (0.000)	0.003*** (0.002)	0.004*** (0.001)
<i>Ln(Tenure)</i>	−0.002* (0.070)	−0.002** (0.046)	−0.003*** (0.009)	−0.003** (0.011)
<i>Control Variables</i>	Yes	Yes	Yes	Yes
<i>Industry Fixed Effects</i>	Yes	No	Yes	No
<i>Year Fixed Effects</i>	Yes	No	Yes	No
<i>Industry × Year Fixed Effects</i>	No	Yes	No	Yes
Observations	12,808	12,808	13,225	13,225
Adjusted R^2	0.530	0.565	0.436	0.466

Panel C: Outsider effect

Definition of outsider	Dependent variable: Total Risk			
	PreCEO Tenure ≤ 4 years		PreCEO Tenure ≤ 1 year	
	(1)	(2)	(3)	(4)
<i>Constant</i>	0.902*** (0.000)	1.009*** (0.000)	0.902*** (0.000)	1.009*** (0.000)
<i>GAI_r</i>	0.021*** (0.000)	0.024*** (0.000)	0.020*** (0.001)	0.023*** (0.000)
<i>GAI_r × Ln(Tenure)</i>	−0.012*** (0.000)	−0.013*** (0.000)	−0.012*** (0.000)	−0.013*** (0.000)
<i>Ln(Tenure)</i>	0.004 (0.214)	0.006* (0.076)	0.004 (0.217)	0.006* (0.079)
<i>Control Variables</i>	Yes	Yes	Yes	Yes
<i>Industry Fixed Effects</i>	Yes	No	Yes	No
<i>Year Fixed Effects</i>	Yes	No	Yes	No
<i>Industry × Year Fixed Effects</i>	No	Yes	No	Yes
Observations	7021	7021	7021	7021
Adjusted R^2	0.626	0.658	0.626	0.658

Panel D: Controlling CEO age (quintile)-by-year fixed effects

	Dependent variable: <i>Total Risk</i>			
	(1)	(2)	(3)	(4)
<i>Constant</i>	0.563*** (0.000)	0.628*** (0.000)	0.568*** (0.000)	0.608*** (0.000)
<i>GAI</i>	0.016*** (0.000)	0.017*** (0.000)	0.015*** (0.000)	0.017*** (0.000)
<i>GAI × Ln(Tenure)</i>	−0.008*** (0.000)	−0.009*** (0.000)	−0.007*** (0.000)	−0.009*** (0.000)
<i>Ln(Tenure)</i>	0.006** (0.016)	0.006** (0.014)	0.005** (0.031)	0.005** (0.024)
<i>Control Variables</i>	Yes	Yes	Yes	Yes
<i>CEO Age × Year Fixed Effects</i>	Yes	Yes	No	No
<i>CEO Age Quintile × Year Fixed Effects</i>	No	No	Yes	Yes
<i>Industry Fixed Effects</i>	Yes	No	Yes	No
<i>Industry × Year Fixed Effects</i>	No	Yes	No	Yes
<i>Observations</i>	12,797	12,797	12,797	12,797
<i>Adjusted R²</i>	0.597	0.626	0.596	0.627

Abbreviations: CEO, chief executive officer; GAI, General Ability Index.

the market model estimated over the window of $[-300, -46]$ days relative to the deal's announcement and the CRSP equally-weighted index return being the proxy for market return. The untabulated results show that generalists are not associated with lower M&A announcement returns in the later stage than in the earlier stage of their tenure.²¹ Collectively, showing that the pattern of tenure weakening the effect of the GAI on firm risk is not more pronounced among weaker governance firm years and that generalists are not more value destroying in the later stage relative to the in earlier stage of their tenure, the above tests help reject the hypotheses that generalists tend to pursue a “quiet life” or “play it safe” as their tenure with the firm increases.

Some easily obtained evidence could further rule out the pursuit of “quiet life” hypothesis. Gormley and Matsa (2016) contend that managers pursuing the “quiet life” would not exert efforts to reduce firm risk (e.g., conduct diversifying M&As). Therefore, the pattern of tenure strengthening the relation between the GAI and the number of diversifying M&As shown in Table 6 rules out the possibility that generalists tend to pursue a more “quiet life” the longer

²¹In this analysis, we regress *CAR5* on *GAI*, *GAI × Ln(Tenure)* and *Ln(Tenure)* of the CEO as of the fiscal year immediately before the announcement, the fixed effects and the control variables, which include acquirer and CEO characteristics (*Ln(CEO Age)*, *Ln(CEO Delta)*, *Ln(CEO Vega)*, *Female Dummy*, *Ln(Assets)*, *Book Leverage*, *Tobin's Q*, *Asset Tangibility*, *R&D Intensity*, *ROA*, *Sales Growth*, *Ln(Firm Age)*, *Free Cash Flow*, *CAPEX* and *Stock Return*) as of the fiscal year immediately before the announcement date and deal characteristics, including *Private Target Dummy*, *Public Target Dummy* and *Diversifying M&A dummy*. The estimated coefficients on *GAI × Ln(Tenure)* are positive and statistically insignificant.

they manage the firms. In addition, Giroud and Mueller (2010) argue that managers pursuing the “quiet life” fail to exert the effort necessary to grow revenues and hold down expenses and, thus, are associated with firms’ lower profitability. We use *ROA* as the measure of firms’ profitability as in Giroud and Mueller (2010) and Gormley and Matsa (2016) and find a strong pattern of tenure strengthening, rather than weakening, the effect of the *GAI* on *ROA* (see Columns 3 and 4 of Table 11 Panel B).

The results we have obtained so far could also help rule out a “performance driven risk-taking” hypothesis. Custódio et al. (2013) note that generalists are high profile and have an easier time being recruited; thus, they could simply have temporary luck but ultimately disappointing performance.²² Perceived poor firm performance decreases the generalist CEOs’ outside job opportunities and thus renders them more risk averse over time. However, in contrast to the premise of this hypothesis, generalists have been shown to be associated with better firm performance (measured as *ROA* or *Risk-Adjusted Stock Return*) in the later stage relative to in the earlier stage of their tenure.

Another alternative explanation for our main findings is that they simply reflect the risk-taking pattern of outside CEOs. Previous studies note that generalists are more likely than specialists to be outsiders and that outsiders are likely to take more risks than insiders particularly in the early years of tenure (e.g., Brockman et al., 2016; Palomino & Peyrache, 2013). Therefore, the observed risk-taking profile of generalists may be attributable to that of outside CEOs. To rule out this potential explanation, we construct a *GAI_r* as the *GAI* that is unexplained by being an outsider and examine the effect of *GAI_r* on *Total Risk* and the evolution of the effect with CEO tenure. Specifically, *GAI_r* is constructed as the residuals from the regressions of the *GAI* on an Outsider dummy, $\ln(\text{Tenure})$, the baseline control variables and fixed effects. In this analysis, outsiders are defined as CEOs with preappointment tenure of not more than 4 years, the median preappointment tenure of our sample, or those with preappointment tenure of not more than 1 year. Then, we run the specifications (2) and (5) of Table 3, with *GAI* being replaced by *GAI_r*. The results reported in Panel C of Table 11 show that the effects of *GAI_r* are very similar as those of *GAI*: The coefficients on *GAI_r* are significantly positive and the coefficients on $\text{GAI_r} \times \ln(\text{Tenure})$ are significantly negative. The results are robust to the alternative definitions of outsiders. Because the effects of the *GAI* remain qualitatively unchanged after removing the component of being an outsider, our main findings do not simply reflect the risk-taking pattern of outsiders.

Lastly, we investigate whether our main findings are driven by the pattern of generalists’ risk-taking behaviour with their age, as CEOs’ tenure and age are closely related and Serfling (2014) documents that older CEOs take on less risk. In this regard, we include in the regressions dummy variables for CEO ages (age groups) to remove the cross-age (age-group) variations. Specifically, we construct CEO age dummies that indicate each CEO age and, by dividing our sample based on the quintile points of the CEO ages, construct CEO age quintile dummies that indicate whether a CEO age belongs to a specific quintile bucket. Then, we interact the CEO age dummies and CEO age quintile dummies, respectively, with year dummies. In Columns (1) and (2) (3 and 4) of Table 11 Panel D, we run similar regressions as specifications (2) and (5) of Table 3, replacing $\ln(\text{CEO Age})$ with $\text{CEO Age} \times \text{Year Fixed Effects}$ ($\text{CEO Age Quintile} \times \text{Year Fixed Effects}$). The results reported in Panel D of Table 11 show a

²²However, the empirical results reported by Custódio et al. (2013) show that CEOs’ general managerial skills have an insignificant effect on firm performance and thus do not support this argument.

risk-taking pattern of generalists that is similar to our main findings; thus, these results indicate that our main findings are not driven by the risk-taking profile of generalists with their age.

6 | CONCLUSIONS

Previous studies argue that, because generalists are easier to switch jobs, their personal wealth is less contingent on the performance and longevity of the current firm that they are leading, and, therefore, they are less risk averse than specialists. Consistent with the notion of generalists being less risk averse, those studies find that generalists are associated with a higher cost of capital of firms and lower survival rates and shorter survival times of newly listed firms and that, moreover, generalists promote innovation. In this paper, we propose the following feature of CEOs' general managerial skills that is absent from the literature: As a CEO stays longer with the same firm, the CEO's skills and knowledge gradually become less general and more firm specific. This feature suggests that any effects of general managerial skills are likely a decreasing function of managerial tenure.

We examine the effect of CEOs' general managerial skills on firm risk and particularly the evolution of the effect with CEO tenure. Using stock return volatility, idiosyncratic risk and operating asset volatility as measures of firm risk, we find a significantly positive effect of CEOs' general managerial skills on firm risk at the early stage of CEO tenure. Then, the effect significantly diminishes with tenure. Furthermore, we find that CEOs' general managerial skills have effects on the riskiness of corporate policies that also present a pattern of weakening with tenure. Further supporting our conjecture of general to firm-specific managerial skills transformation, we show that our main findings are more pronounced among outside CEOs than among inside CEOs.

Our study addresses an important dimension of general managerial skills that has not been reported in the literature, namely, CEO tenure. This dimension suggests that generalist CEOs staying with the same firm are bound to become specialists at a certain stage, and, thus, their behaviour and impact on firms are expected to vary with their tenure. On the basis of this, to obtain a clear understanding of the impact of CEOs' general managerial skills, we strongly suggest that future studies consider the role of CEO tenure in the evolution of this managerial trait.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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APPENDIX

TABLE A1 Variable definitions and data resources

CEO characteristics	
<i>GAI</i>	An index constructed by Custódio et al. (2013) as a measure of a CEO's general ability. It is measured by the first principal component of (1) the number of different positions held by the CEO, (2) the CEO's employment at a number of different firms, (3) the CEO's employment in a number of different industries classified at four-digit SIC codes, (4) a binary variable taking a value of 1 if the CEO held a CEO position at another firm before her/his current position and (5) a binary variable taking a value of 1 for multidivision (Custódio et al., 2013).
<i>Generalist Dummy</i>	A dummy variable that takes the value of one if <i>GAI</i> is above the yearly median, and zero otherwise (Custódio et al., 2013).
<i>Tenure</i>	The number of years since assuming the CEO position (Compustat ExecuComp).
<i>CEO Age</i>	The CEO's age (Compustat ExecuComp).
<i>CEO Delta</i>	Dollar change in the portfolio value for a 1% change in the stock price (in \$ thousands) (Coles et al., 2006; Core & Guay, 2002).
<i>CEO Vega</i>	Dollar change in the portfolio value for a 0.01 change in the annualized standard deviation of stock returns (in \$ thousands) (Coles et al., 2006; Core & Guay, 2002).
<i>Female Dummy</i>	A dummy variable that takes the value of one if the CEO is a female, and zero otherwise (Compustat ExecuComp).
Firm characteristics	
<i>Total Risk</i>	The annualized standard deviation of daily stock returns over the fiscal year (CRSP).
<i>Idiosyncratic Risk</i>	The annualized standard deviation of the residuals from the regressions of daily stock return on Fama–French three factors estimated over the fiscal year (CRSP; Kenneth R. French's website https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html).
<i>Operating Assets Volatility</i>	<i>Total Risk</i> times the ratio of the firm's market value to enterprise value (CRSP; Compustat $Total Risk \times (CSHO \times PRCC_F) / [LT + (CSHO \times PRCC_F) - CH]$).
<i>R&D Intensity</i>	Research and development expenses divided by total assets (Compustat XRD/AT).
<i>Patents</i>	The number of patents filed by a firm in a fiscal year (Extended data set of KPSS, 2017).
<i>Book Leverage</i>	Total debt, defined as long-term debt plus debt in current liabilities, divided by total assets [Compustat $(DLTT + DLC)/AT$].
<i>Operating Leverage</i>	The percentage change in operating income for a percentage increase in sales and is estimated using quarterly data from year t to year $t + 2$ (Compustat).

<i>Segments</i>	The number of business segments (Compustat).
<i>Total Assets (\$ millions)</i>	(Compustat AT).
<i>Tobin's Q</i>	Market value of total assets divided by book value of total assets [Compustat $(AT - CEQ + PRCC_F \times CSHO)/AT$].
<i>Asset Tangibility</i>	Net property, plant and equipment divided by total assets (Compustat PPENT/AT).
<i>ROA</i>	Earnings before interest and taxes divided by total assets (Compustat EBIT/AT).
<i>Sales Growth</i>	The growth rate of sales [Compustat $(SALE_t/SALE_{t-1}) - 1$].
<i>Firm Age</i>	The number of years since the firm was listed in CRSP for the first time (CRSP).
<i>CAPEX</i>	Capital expenditure divided by total assets (Compustat CAPX/AT).
<i>Labour Stock</i>	The number of employees divided by total assets (Compustat EMP/AT).
<i>Free Cash Flow</i>	Free cash flow divided by total assets [Compustat $(OIBDP - XINT - TXT - CAPX)/AT$].
<i>Stock Return</i>	Stock return over the fiscal year (CRSP).
<i>Risk-Adjusted Stock Return</i>	Stock return adjusted by the Fama–French three-factor model over the fiscal year (CRSP; Kenneth R. French's website https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html).
<i>G Index</i>	Governance index of Gompers et al. (2003), which is based on 24 antitakeover provisions (ISS).
M&A variables	
<i>Total No. of M&As</i>	The total number of M&As announced in a fiscal year (SDC).
<i>No. of Diversifying M&As</i>	The number of diversifying M&As announced in a fiscal year (SDC).
<i>CAR5</i>	The acquirer's percentage cumulative abnormal return over $[-2, 2]$ days around the deal's announcement (SDC; CRSP).
<i>Private Target Dummy</i>	A dummy variable that takes the value of one if the target is a private firm, and zero otherwise (SDC).
<i>Public Target Dummy</i>	A dummy variable that takes the value of one if the target is a public firm, and zero otherwise (SDC).
<i>Diversifying M&A dummy</i>	A dummy variable that takes the value of one if the deal's acquirer and target do not share the same three-digit SIC code, and zero otherwise (SDC).